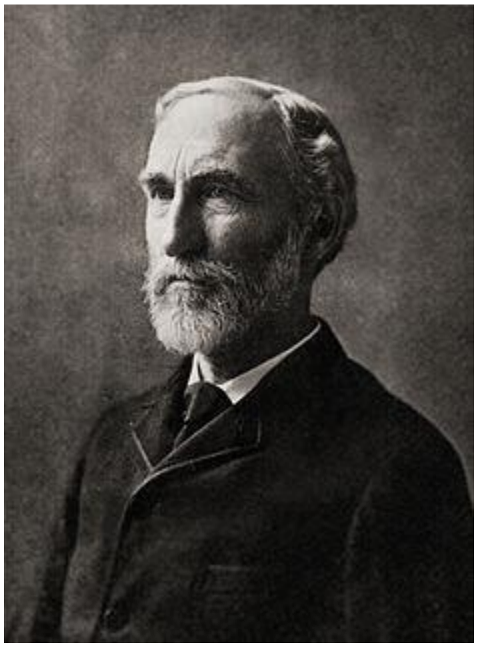
Chapter 6. Phase diagrams

Outline

- Phases, components, and degrees of freedom
  - Definitions
  - The Gibbs phase rule
- Two-component system
  - Vapour pressure diagrams
  - Temperature-composition diagrams
  - Liquid-liquid phase diagrams
  - Liquid-solid phase diagrams

6.1 Phases, components, and degrees of freedom



Josiah Willard Gibbs
(1839 –1903)

J. W. Gibbs was an American scientist who made important theoretical contributions to physics, chemistry, and mathematics. His work on the applications of thermodynamics was instrumental in transforming physical chemistry into a rigorous deductive science. Together with James Clerk **Maxwell** and Ludwig **Boltzmann**, he created statistical mechanics (a term that he coined), explaining the laws of thermodynamics as consequences of the statistical properties of large ensembles of particles.

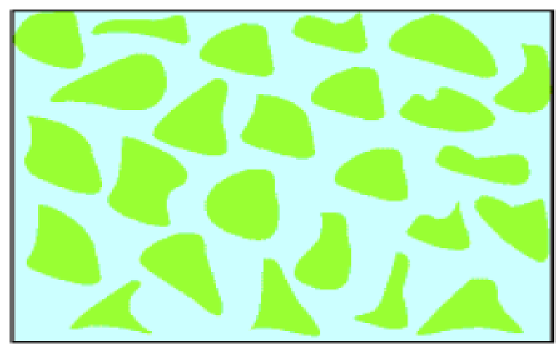
1. Definitions

(a) Phases (P)

A phase of a substance is a form of matter that is uniform throughout in chemical composition and physical state.

(a)

A single-phase solution, in which the composition is uniform on a microscopic scale.



(b)

A dispersion, in which regions of one component are embedded in a matrix of a second component.

The number of phases in a system is denoted P.

- Ice is a single phase even though it might be chipped into small fragments.
- A crystal is a single phase.
- Two totally miscible liquids form a single phase.
- A gas, or a gaseous mixture, is a single phase.

P=1

Any sample cut from the sample, however small, is representative of the composition of the whole.

- A slurry of ice and water is a two-phase system.
- An alloy of two metals is a two-phase system if the metals are immiscible.
- A dispersion of clays in water is uniform on a macroscopic scale but not on a microscopic scale.

P=2

A small sample comes entirely from one of the minute grains of pure A and would not be representative of the whole.

(b) Components (C) and species (S)

A constituent is a chemical species that is present in a system (S).

A component is a chemically independent constituent of a system (C).

The number of components, C, in a system is the minimum number of independent species necessary to define the composition of all the phases present in the system.

Usually, $C \leq S$

- When the species do not react, $C = S$

- If the species do react, and are at equilibrium,
  $C = S - R$
  R---the number of independent reactions

- The limited concentration in the same phase
  $C = S - R - R'$
  R'---the number of concentration limitation
  (in one phase)

$$H_2O \rightleftharpoons H^+ + OH^-$$

How many species? $S = 3$

How many independent reactions? $R = 1$

How many concentration limitations? $R' = 1$

How many components? $C = S - R - R' = 1$

Note: S can be different according to your opinion, but C is only one for a given system.

$$CaCO_3\,(s) \rightleftharpoons CaO\,(s) + CO_2\,(g)$$

How many phases? $P = 3$

How many species? $S = 3$

How many independent reactions? $R = 1$

How many concentration limitations? $R' = 0$
(in one phase)

How many components? $C = S - R - R' = 2$

Example

How many components are present in a system in which ammonium chloride undergoes thermal decomposition?

Method Begin by writing down the chemical equation for the reaction and identifying the constituents of the system (all the species present) and the phases. Then decide whether, under the conditions prevailing in the system, any of the constituents can be prepared from any of the other constituents. The removal of these constituents leaves the number of independent constituents. Finally, identify the minimum number of these independent constituents that are needed to specify the composition of all the phases.

$$NH_4Cl\,(s) \rightleftharpoons NH_3\,(g) + HCl\,(g)$$

How many phases? $P = 2$

How many species? $S = 3$

How many independent reactions? $R = 1$

How many concentration limitations? (in one phase) $R' = 1$

How many components? $C = S - R - R' = 1$

(c) Degrees of freedom (F)

Degrees of freedom ,or the variance of a system, F, is the number of intensive variables that can be changed independently without disturbing the number of phases in equilibrium. In a single-component, single-phase system (C= 1, P= 1), p and T may be changed independently without changing the number of phases, so F = 2. We say that such a system is bi-variant, or that it has two degrees of freedom.

2. The phase rule

$$F = C - P + 2$$

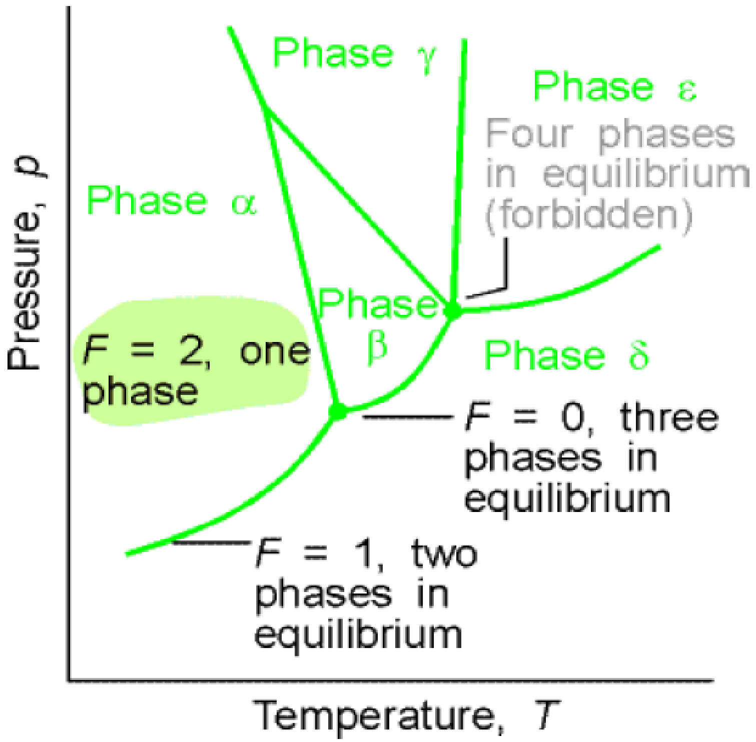
J.W. Gibbs deduced the phase rule:

F: the degree of freedom or variance;

C: the number of components;

P: the number of phases at equilibrium.

One-component system



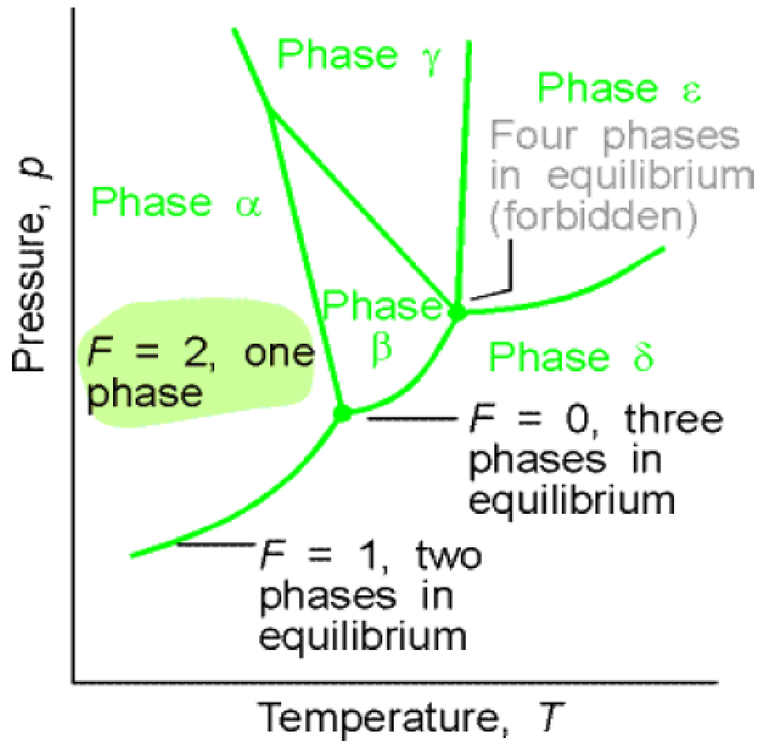
The typical regions of a one-component phase diagram.

If only one phase is present:

$$F = 1 - 1 + 2 = 2$$

Both p and T can be varied independently without changing the number of phases. In other words, a single phase is represented by <u>an area</u> on a phase diagram.

One-component system



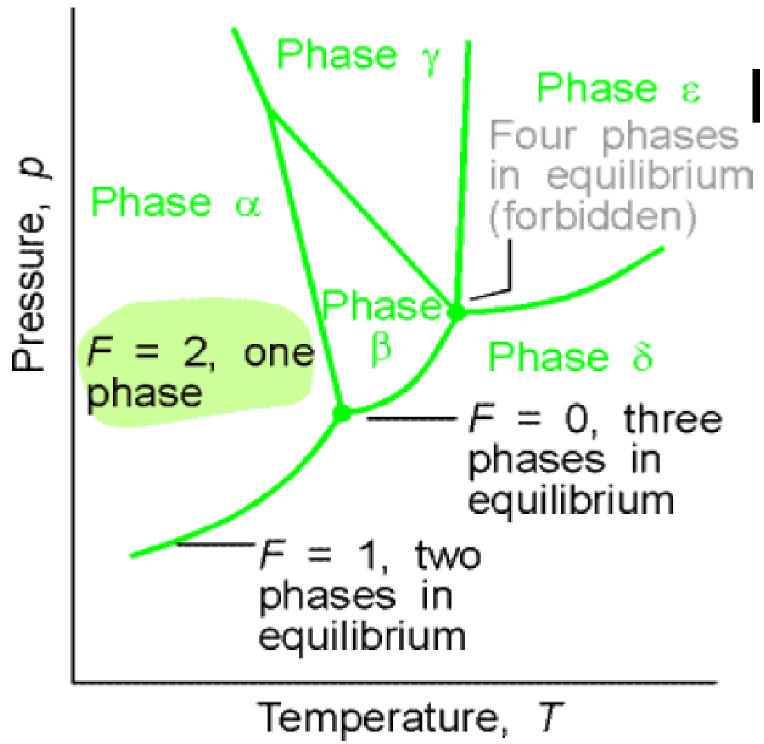
The typical regions of a one-component phase diagram.

If two phases are in equilibrium:

$$F = 1-2+2=1$$

It implies that p is not freely variable if the T is set. The equilibrium of two phases is represented by a line in the phase diagram. Therefore, freezing (or any other phase transition) occurs at a definite T at a given p.

One-component system



The typical regions of a one-component phase diagram.

If three phases are in equilibrium:

$$F = 1-3+2=0$$

The system is invariant. The special condition can be established only at the definite T and p that is characteristic of the substance and outside our control. The equilibrium of three phases is therefore represented by a point, the triple point, on the phase diagram

In a one-component system:

Only one phase, an area:

$C = 1, P = 1 \Rightarrow F = 1 - 1 + 2 = 2$

Two phases, The lines:

$C = 1, P = 2 \Rightarrow F = 1 - 2 + 2 = 1$

three phases, A point:

$C = 1, P = 3 \Rightarrow F = 1 - 3 + 2 = 0$

Four phases cannot mutually coexist in equilibrium.

6.2 Two component ideal systems

If $C = 2$ $\qquad F = C - P + 2 = 4 - P$

$$\boldsymbol{F = 4 - P}$$

$$\because \boldsymbol{P_{min} = 1} \qquad \therefore \boldsymbol{F_{max} = 3}$$

①temperature, ②pressure, ③composition

If T or p is kept constant, the remaining variance is:

Now $P_{min} = 1 \quad F_{max} = 2$

Hence,

(a) One form of the phase diagram is a map of p and composition (in this case, T could be held constant)

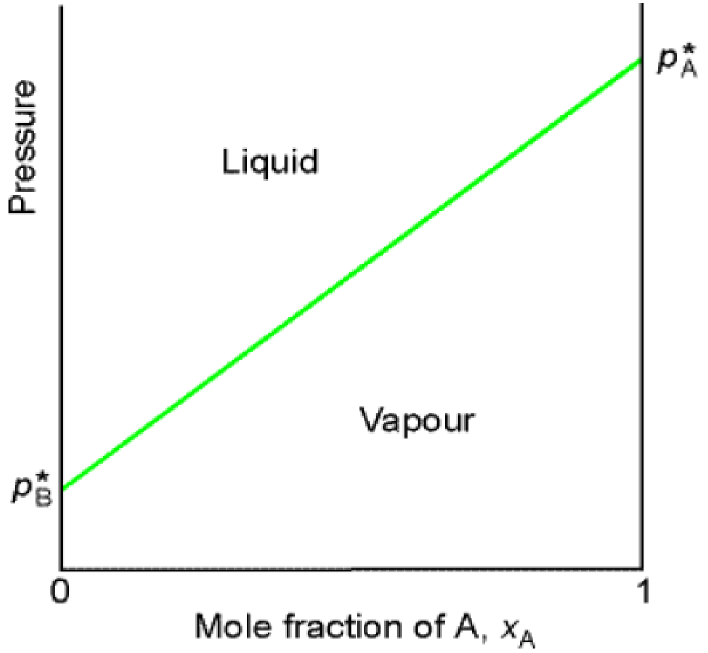
(b) The other phase diagram depicted in terms of T and composition.(P could be held constant)

1. Vapour pressure diagrams

An ideal solution of two volatile liquids:

$$p_A = x_A\, p_A{}^* \qquad\qquad p_B = x_B p_B{}^*$$

The total vapor pressure of the mixture is:



$$p = p_A + p_B = x_A p_A{}^* + x_B p_B{}^*$$
$$= p_B{}^* + (p_A{}^* - p_B{}^*)x_A$$

$$x_A = 0 \quad \rightarrow \quad p = p_B{}^*$$

$$x_A = 1 \quad \rightarrow \quad p = p_A{}^*$$

x_A & x_B: The mole fractions of A & B in the liquid phase.

y_A & y_B: The mole fractions of A & B in the gas phase.

It is possible and common:

$$\boldsymbol{y_A \neq x_A} \qquad\qquad \boldsymbol{y_B \neq x_B}$$

The compositions of the liquid and vapour that are in mutual equilibrium are not necessarily the same

According to Dalton's law

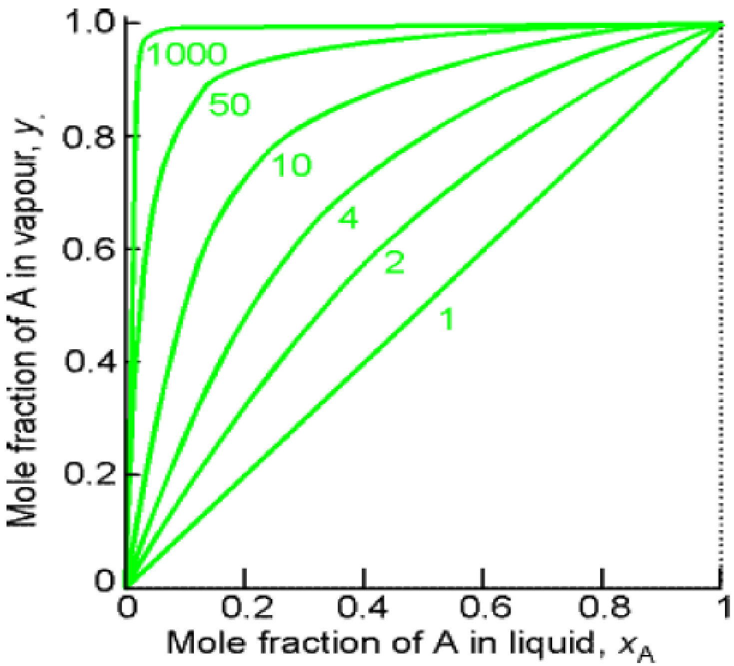
$$y_A = \frac{p_A}{p} \qquad y_B = \frac{p_B}{p}$$

$$\begin{cases} p_A = x_A\, p_A{}^* \\ p = p_B{}^* + (p_A{}^* - p_B{}^*)x_A \end{cases}$$

$$y_A = \frac{x_A\, p_A{}^*}{p_B{}^* + (p_A{}^* - p_B{}^*)x_A} \qquad y_B = 1 - y_A$$

$$x_A = f(y_A) \Rightarrow \boxed{p = p_B{}^* + (p_A{}^* - p_B{}^*)x_A}$$

$$\boxed{p = \frac{p_A{}^* p_B{}^*}{p_A{}^* + (p_B{}^* - p_A{}^*)y_A}}$$



The mole fraction of A in the vapour of a binary ideal solution expressed in terms of its mole fraction in the liquid for various values of p^*_A/p^*_B with A more volatile than B. In all cases the vapour is richer than the liquid in A.

$$\frac{y_B}{y_A} = \frac{p_B}{p_A} = \frac{x_B\, p_B{}^*}{x_A\, p_A{}^*}$$

If A is more volatile than B

$$p_A{}^* > p_B{}^* \qquad \frac{y_B}{y_A} < \frac{x_B}{x_A}$$

$$\frac{1-y_A}{y_A} < \frac{1-x_A}{x_A} \qquad \frac{1}{y_A} < \frac{1}{x_A}$$

$$y_A > x_A$$

The vapor is richer than the liquid in the more volatile component.

① The liquid line

$$\boxed{p = p_B{}^* + (p_A{}^* - p_B{}^*)x_A}$$

$$x_A = 0 \quad \rightarrow \quad p = p_B{}^*$$
$$x_A = 1 \quad \rightarrow \quad p = p_A{}^*$$

② The vapour line

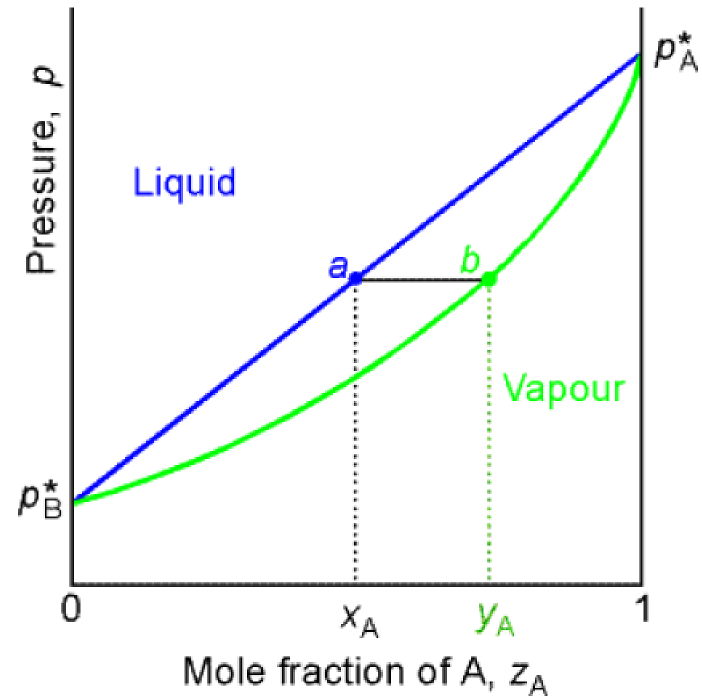
$$\boxed{p = \frac{p_A{}^* p_B{}^*}{p_A{}^* + (p_B{}^* - p_A{}^*)y_A}}$$

$$y_A = 0 \quad \rightarrow \quad p = p_B{}^*$$
$$y_A = 1 \quad \rightarrow \quad p = p_A{}^*$$

③ If $p_A{}^* > p_B{}^*$

$y_A > x_A$

The pressure-composition diagrams

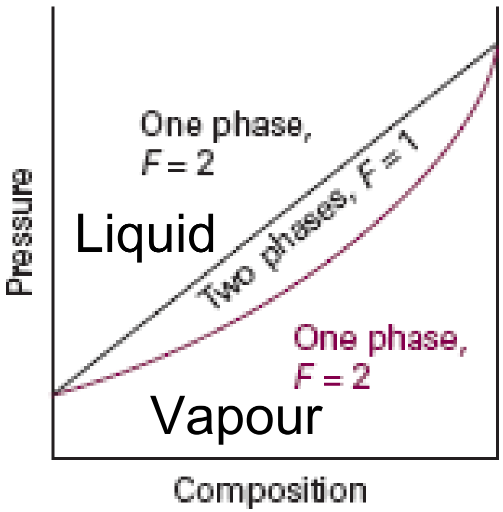


The dependence of the total vapour pressure of an ideal solution on the mole fraction of A in the entire system.

2.The interpretation of the diagrams

p-composition diagram: at constant T, one variance is discarded:

$$F' = F - 1$$



- Above the liquid line is the liquid phase.
- below the vapour line is the vapour phase.

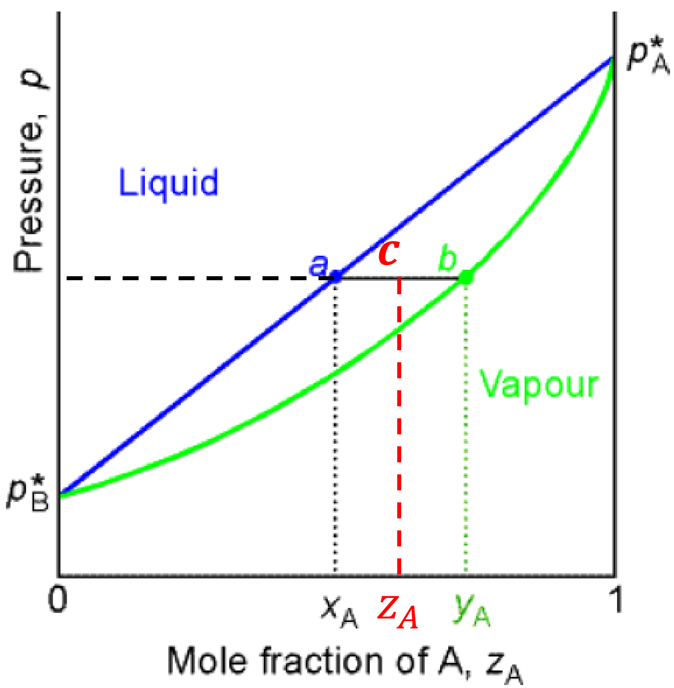
  P=1 F=2−1+2=3 F'=3−1=2

  p and (x_A or y_A) can be varied independently

- Between the liquid line and the vapour line: liquid-vapour co-existence.

  P=2 F=2−2+2=2 F'=2−1=1

  For a give p, the variance is 0, and the vapour and liquid phases have fixed compositions.

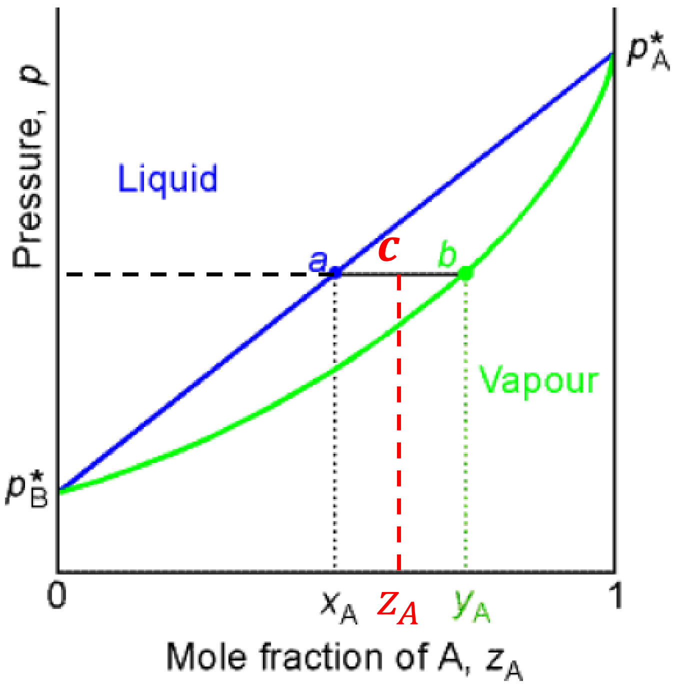


- **Phase point:**

a: vapour pressure of a mixture of composition x_A.
b: the composition of the vapour, y_A, that is in equilibrium with the liquid.

- **Tie line:**

A tie-line is a horizontal (i.e., constant-p or T) line through the chosen point, which intersects the phase boundary lines on either side.



- On the tie line:

As p is constant, the variance is 0. The composition (x_A y_A) is constant through the line.

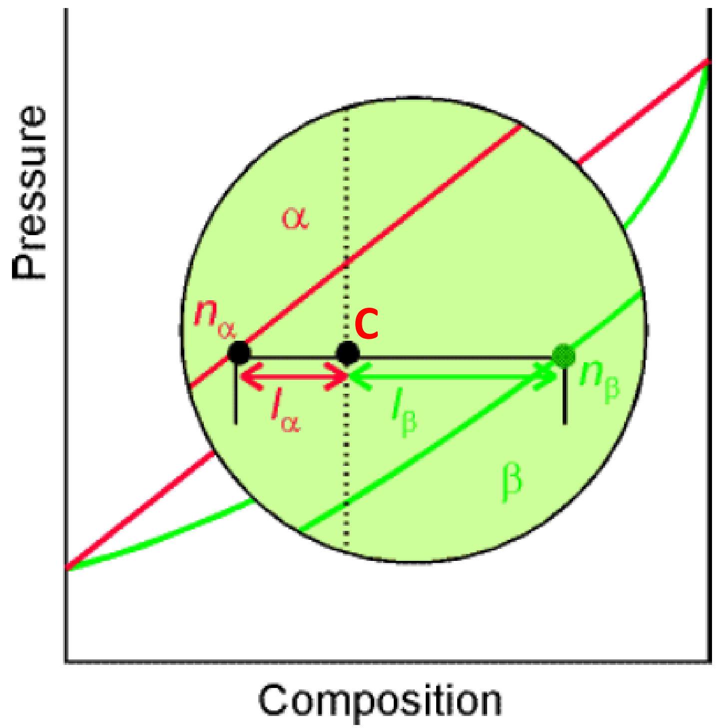
c: the overall composition z_A

$$z_A = \frac{n_A}{n_A + n_B} = \frac{x_A n_l + y_A n_g}{n_l + n_g}$$

$$z_A(n_l + n_g) = x_A n_l + y_A n_g$$

$$\frac{n_l}{n_g} = \frac{y_A - z_A}{z_A - x_A} = \frac{bc}{ca}$$

- ## The level rule:

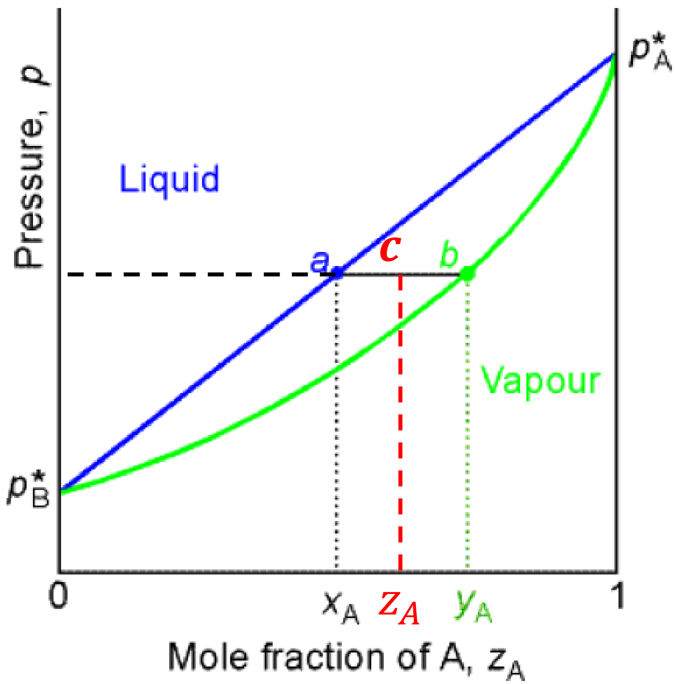


A point in the two-phase region of a phase diagram indicates not only qualitatively that both liquid and vapour are present, but represents quantitatively the relative amounts of each.

Two phases α and β are co-existence in equilibrium.

The distances from point c to the phase points along the tie line are l_α and l_β:

$$n_\alpha l_\alpha = n_\beta l_\beta$$ The level rule



- **Point a:**

$$\frac{l_{vap}}{l_{liq}} \approx infinite\ large \qquad \frac{n_{liq}}{n_{vap}} \approx infinite\ large$$

Trace amount of vapour phase (*same* y_A as b)

- **Point c: if ac=cb**

$$\frac{l_{vap}}{l_{liq}} = 1 \qquad \frac{n_{liq}}{n_{vap}} = 1$$

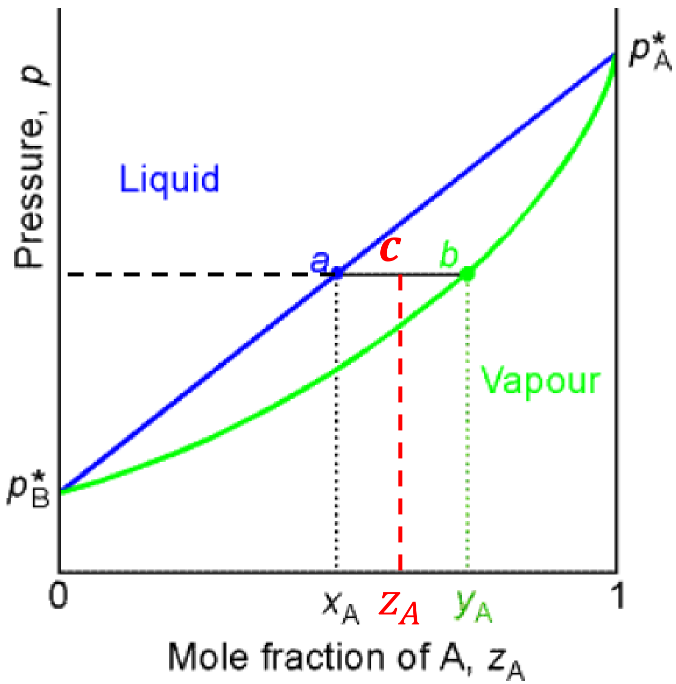
Equal amounts of liquid and vapour phase

- **Point b:**

$$\frac{l_{vap}}{l_{liq}} \approx infinite\ small \qquad \frac{n_{liq}}{n_{vap}} \approx infinite\ small$$

Trace amount of liquid phase (*same* x_A as a)

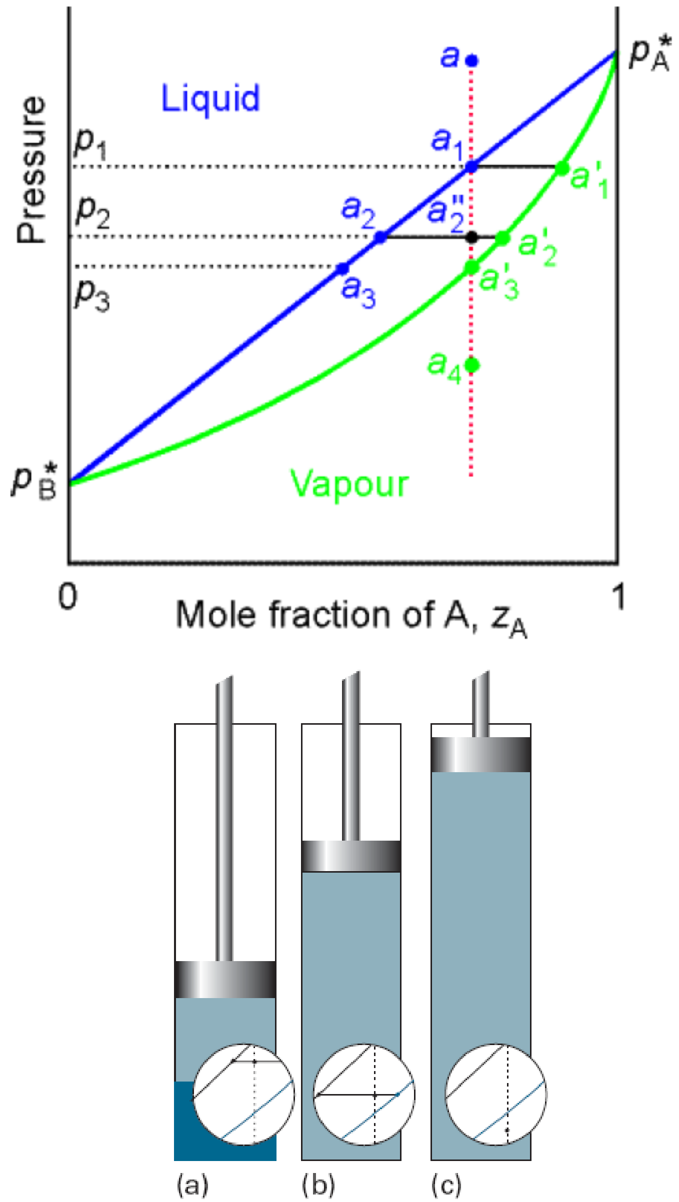
However, in spite of any change in amount, the liquid composition (x_A) and the vapour composition (y_A) are always constant on the tie line.



At given P

1) All the points on the tie line have the same composition (x_A, y_A).

2) On the tie line, $\frac{n_l}{n_g} = \frac{y_A - z_A}{z_A - x_A} = \frac{bc}{ca}$ changes according to the level rule, as a result, the $z_A = \frac{n_A}{n_A + n_B} = \frac{x_A n_l + y_A n_g}{n_l + n_g}$ (the overall composition) changes.



- The isopleth line等浓度线:

a: pure liquid phase;

a_1: evaporation begins;

trace amount of vapour phase;

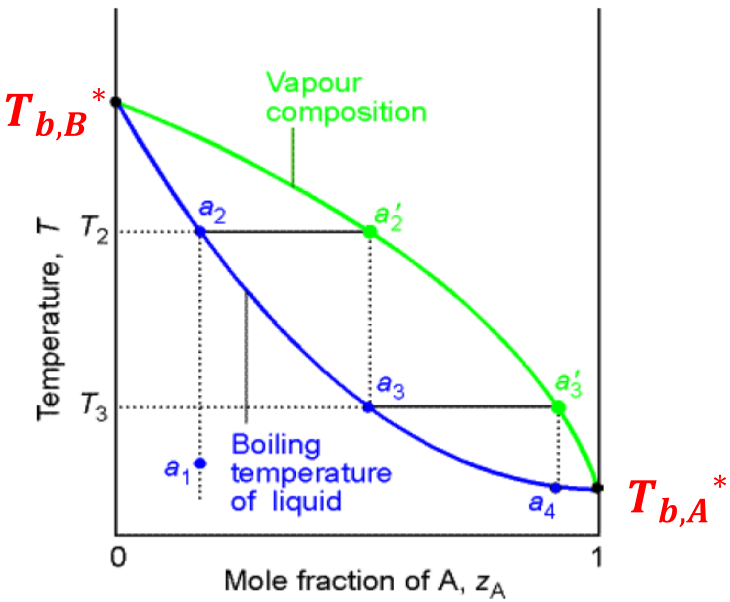
a_2'': two phases in equilibrium;

$n_{vap} > n_{liq}$

a_3': trace amount of liquid phase;

a_1: pure vapour phase.

3. The temperature-composition diagrams



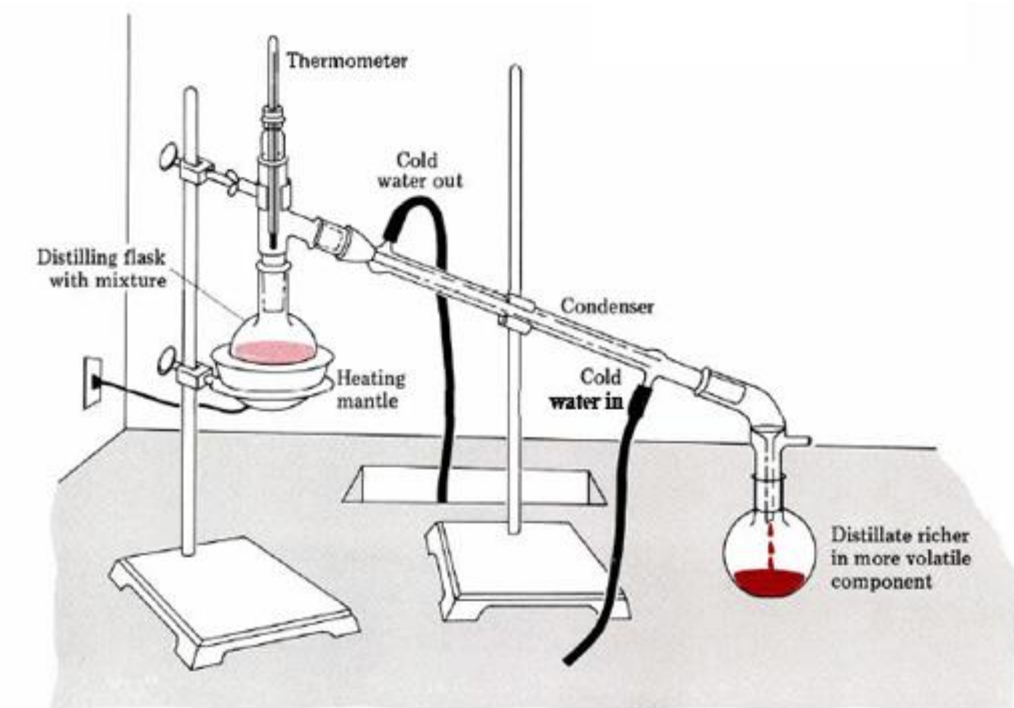
At constant p

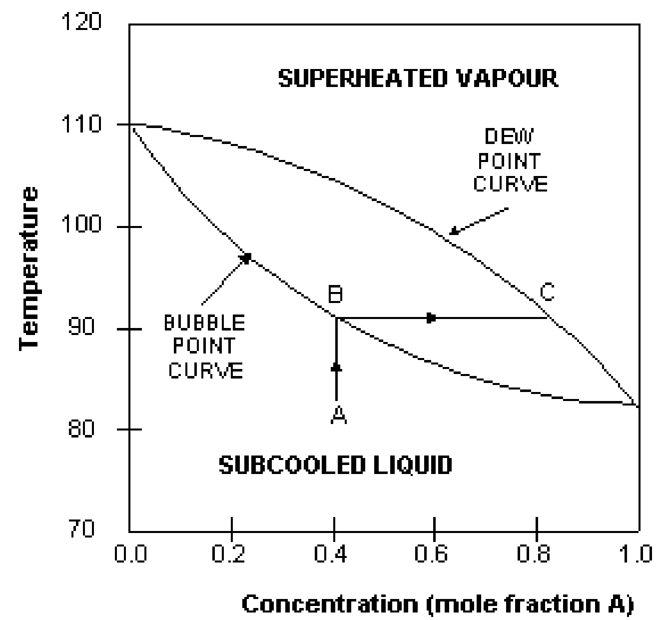
$f(T, x_A, y_A)$

- Below the liquid line is the liquid phase.
- Above the vapour line is the vapour phase.
- Between the liquid line and the vapour line: liquid-vapour co-existence.
- $T_{b,A}^* < T_{b,B}^*$ A is more volatile.
  $y_A > x_A$

4. The distillation of mixtures

Simple distillation





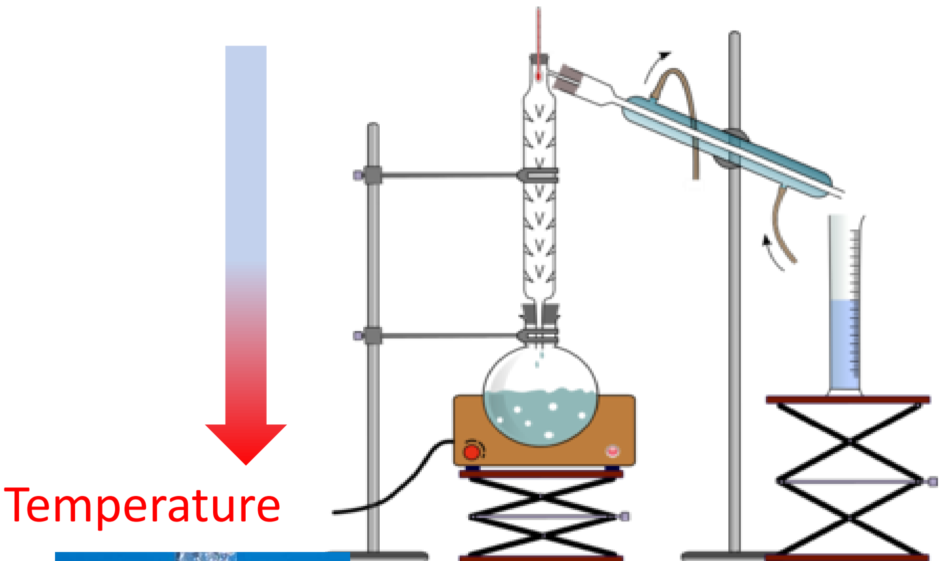
$y_A > x_A$

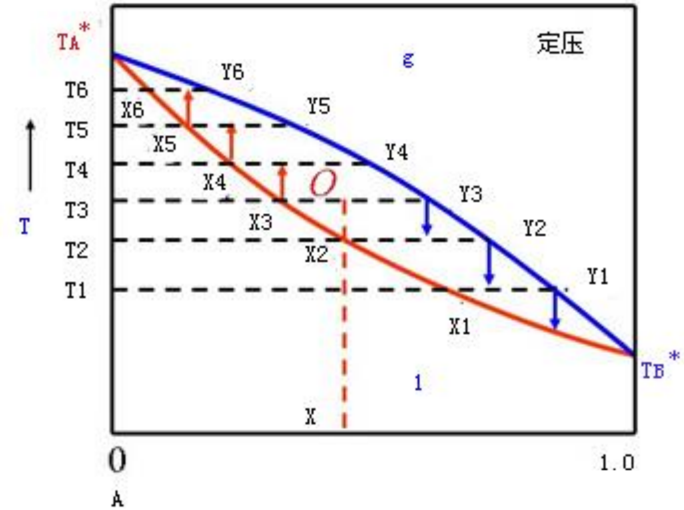
A is richer in the vapour phase than in the liquid phase; B is richer in the remaining liquid phase.

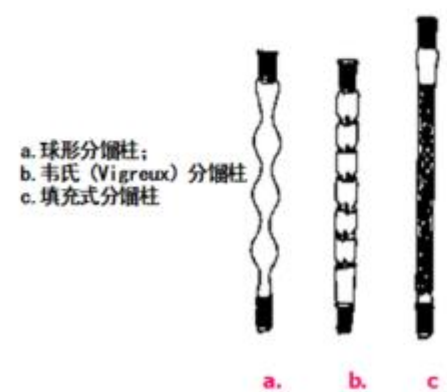
Fractional distillation

The boiling and condensation cycle is repeated successively.

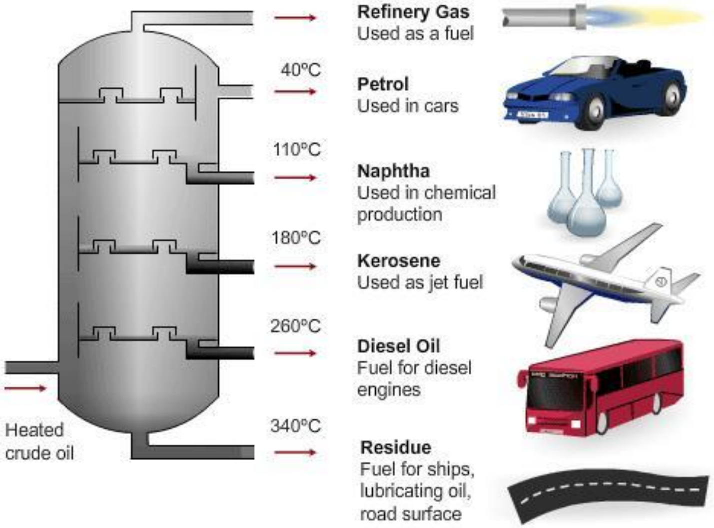
A typical lab fractional distillation would look like this:

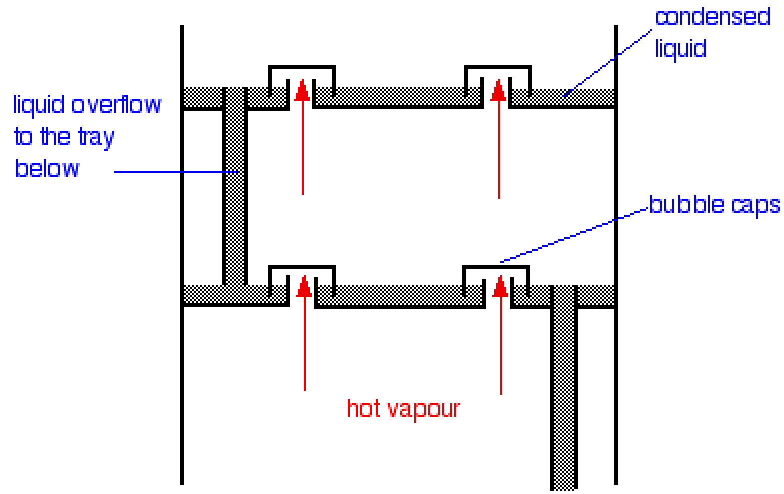


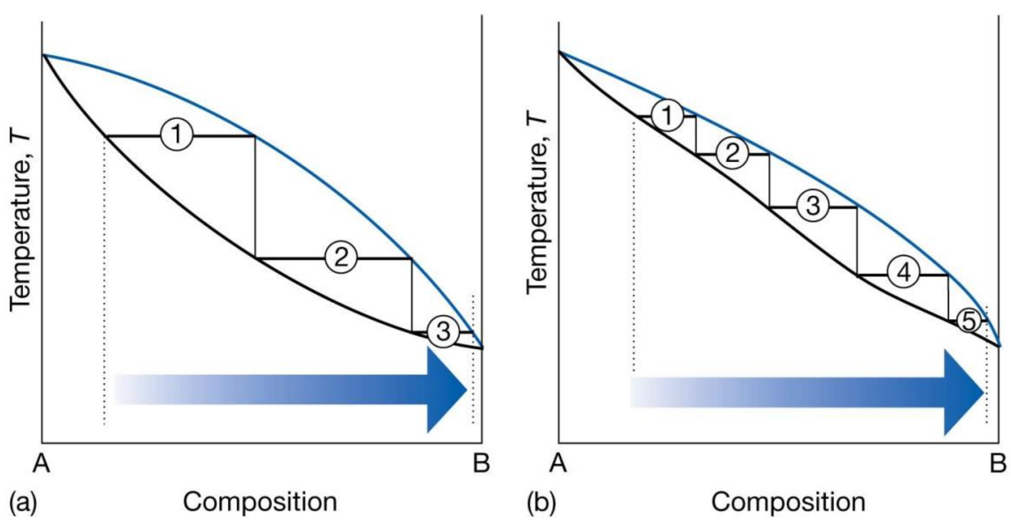




Fractional distillation





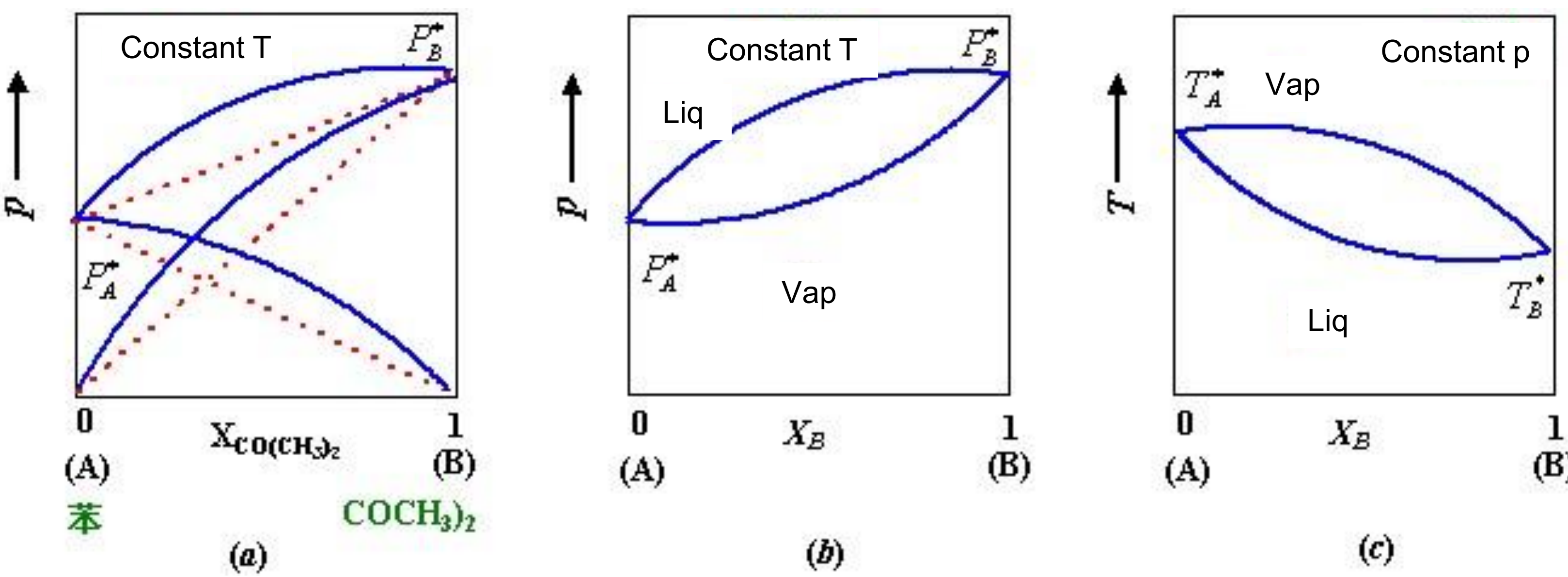


The efficiency of a fractionating column is expressed in terms of the number of theoretical plates, the number of effective vaporization and condensation steps that are required to achieve a condensate of given composition from a given distillate.

6.3 Two component real systems

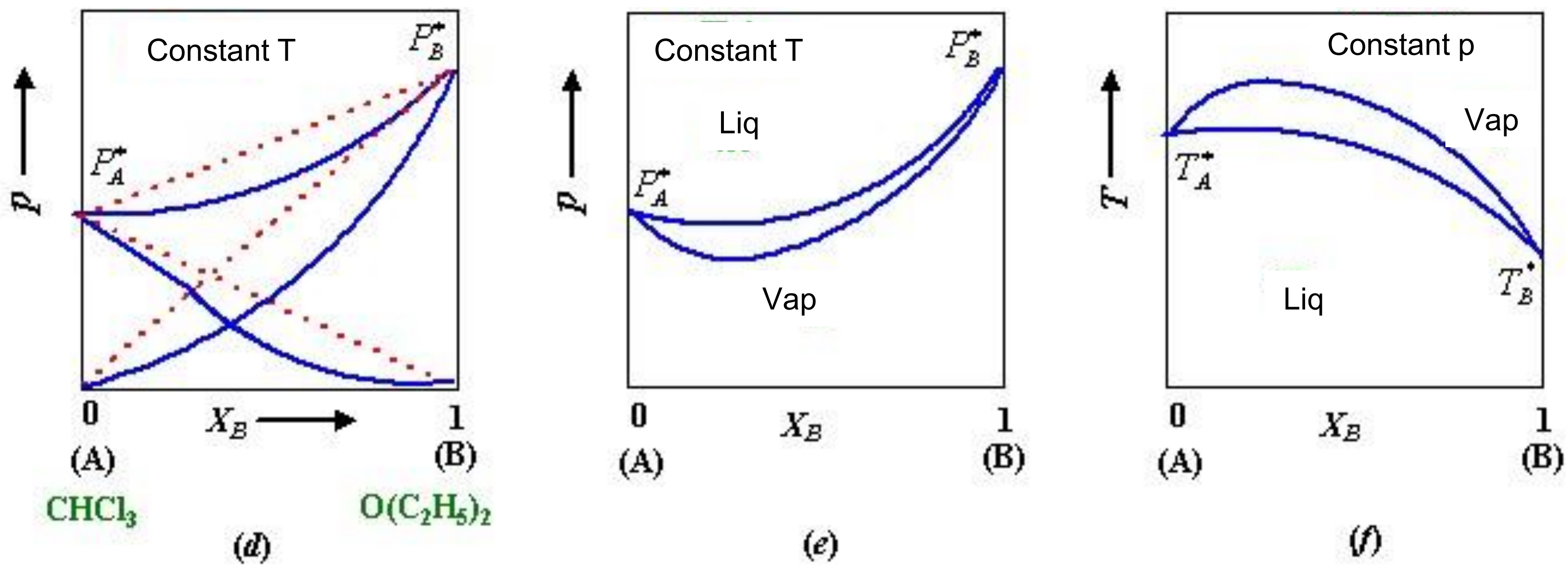
Positive deviation from Raoult's Law

$$P_A > P_A{}^* x_A \qquad\qquad P_B > P_B{}^* x_B$$



Negative deviation from Raoult's Law

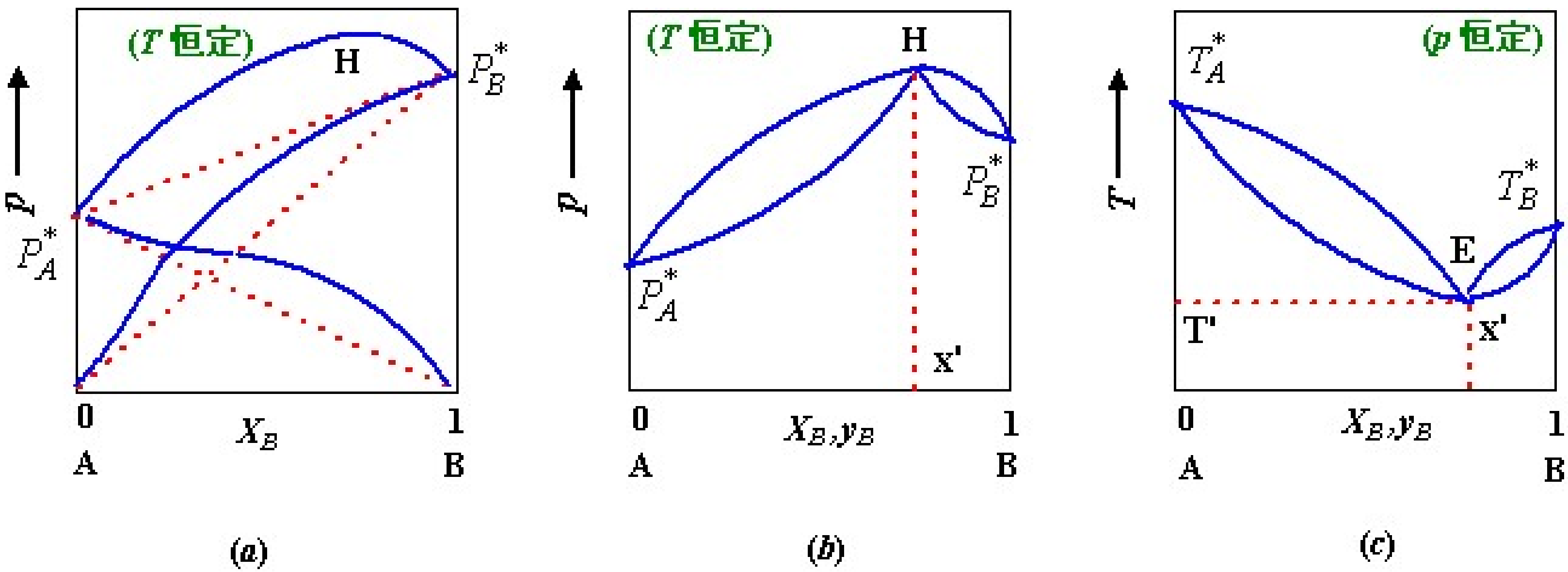
$$P_A < {P_A}^{*} x_A \qquad P_B < {P_B}^{*} x_B$$



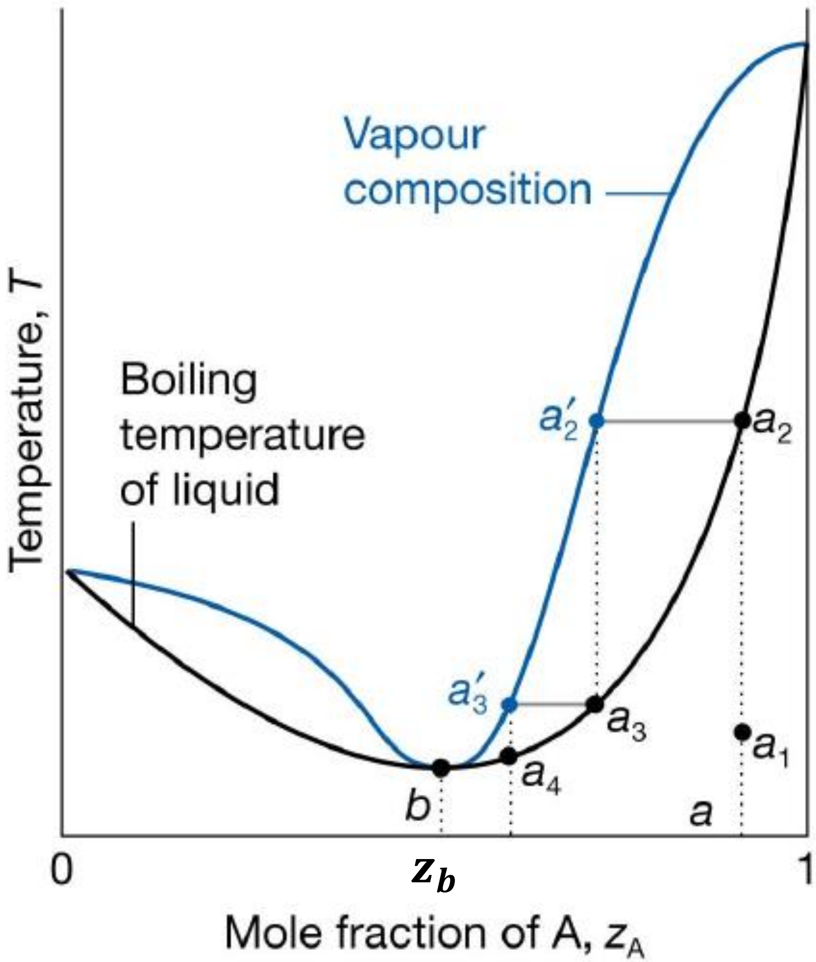
Significant positive deviation

$$P_{real} \gg P_B{}^*$$

A maximum　　A maximum　　A minimum



The mixture is destabilized relative to the ideal solution, the A–B interactions then being unfavourable. For such mixtures G^E is positive (less favourable to mixing than ideal), and there may be contributions from both enthalpy and entropy effects. Examples include dioxane/water and ethanol/water mixtures.



Point b:
minimum azeotropic恒沸点 point

At b, the vapour has the same composition as the liquid. Evaporation then occurs without change of composition.

The mixture at b is called azeotrope.

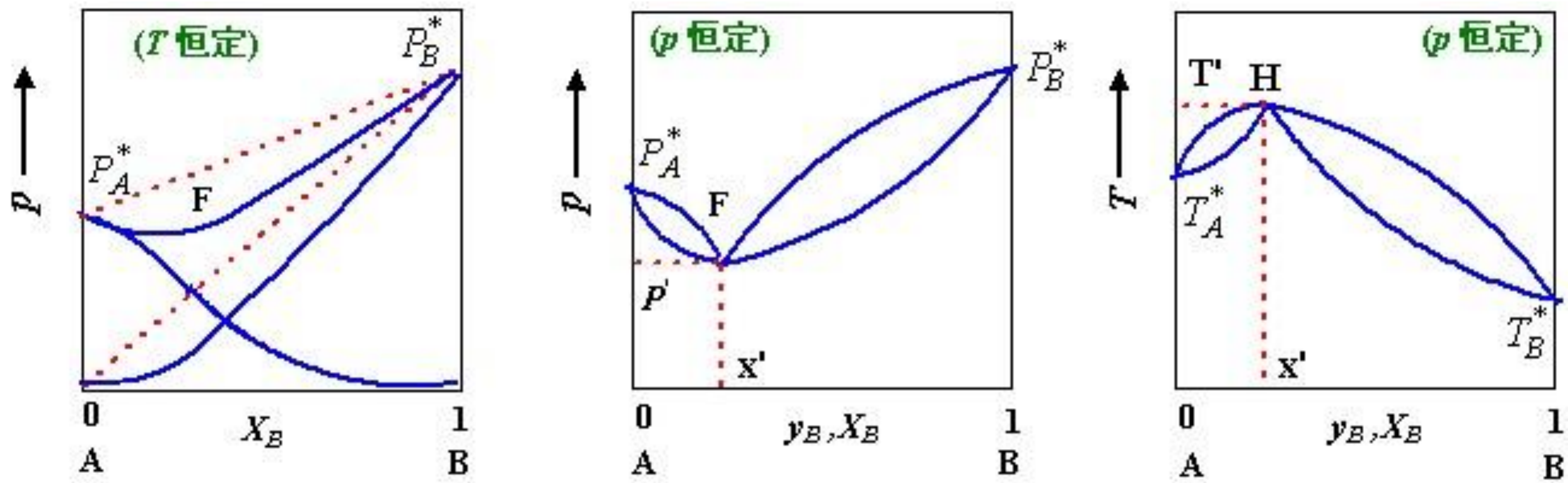
If $z_A > z_b$ Pure A Azeotrope (z_b)

If $z_A < z_b$ Pure B Azeotrope (z_b)

Significant negative deviation

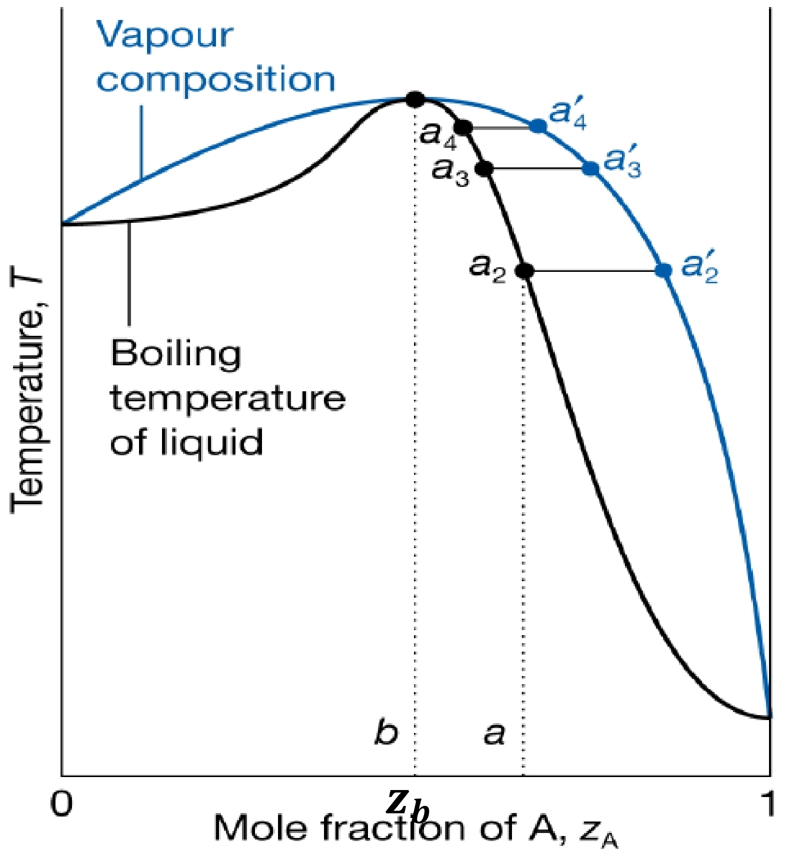
$$P_{real} \ll P_A{}^*$$

A minimum A minimum A maximum



(a) (b) (c)

A maximum in the phase diagram may occur when the favourable interactions between A and B molecules reduce the vapour pressure of the mixture below the ideal value: in effect, the A–B interactions stabilize the liquid. In such cases the excess Gibbs energy, G^E, is negative (more favourable to mixing than ideal). Examples of this behaviour include trichloromethane/propanone and nitric acid/water mixtures.



Point b: maximum azeotropic point

At b, the vapour has the same composition as the liquid. Evaporation then occurs without change of composition.

The mixture at b is called azeotrope.

If $z_A > z_b$ Pure A
Azeotrope (z_b)

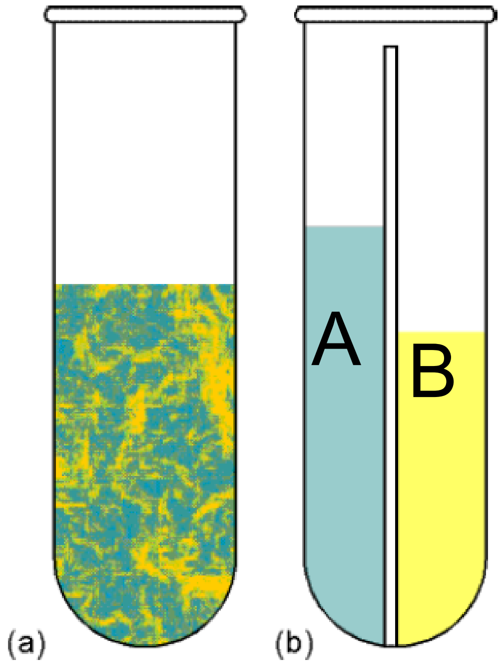
If $z_A < z_b$ Pure B
Azeotrope (z_b)

6.4 Immiscible liquids

$$p = p_A{}^* + p_B{}^*$$

$$T_{b,mix} < T_{b,A}{}^*$$

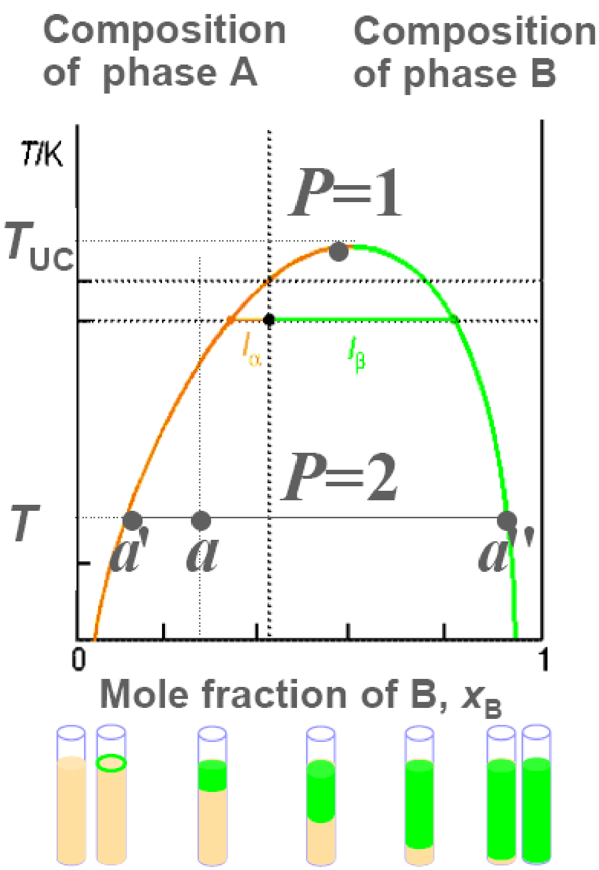
$$T_{b,mix} < T_{b,B}{}^*$$



The distillation of (a) two immiscible liquids can be regarded as (b) the joint distillation of the separated components,

The presence of the saturated solutions means that the 'mixture' boils at a lower temperature than either component would alone because boiling begins when the total vapour pressure reaches 1 atm, not when either vapour pressure reaches 1 atm. This distinction is the basis of **steam distillation**, which enables some heat-sensitive, water-insoluble organic compounds to be distilled at a lower temperature than their normal boiling point.

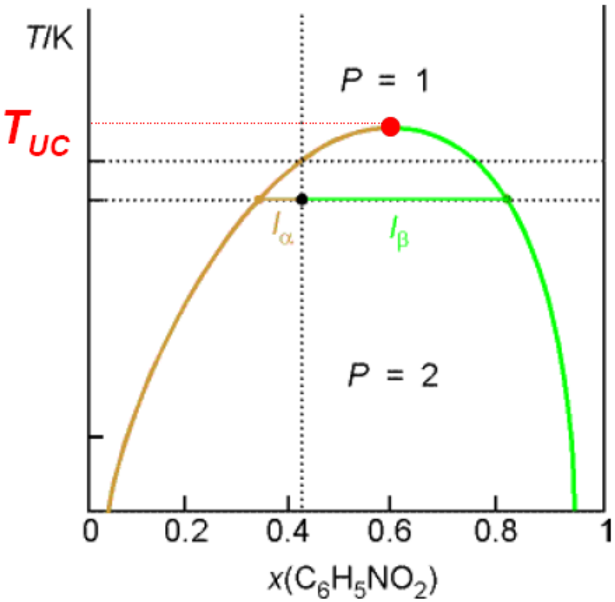
6.5 Partially miscible liquid-liquid diagrams



The *T-C* diagram for hexane and nitrobenzene at 1 atm

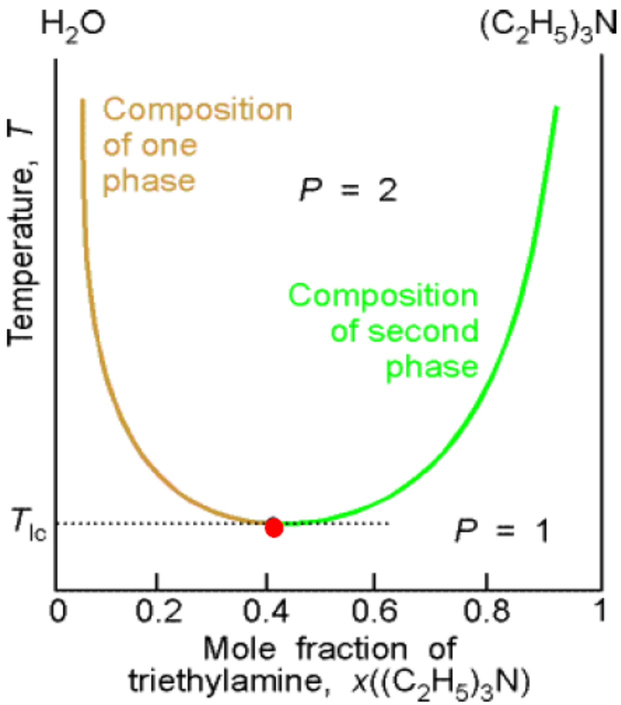
- Let a small amount of a liquid B be added to a sample of another liquid A at T. It dissolves completely, and the binary system remains a single phase. (before a')
- As more B is added, a stage comes at which no more dissolves. The sample now consists of two phases in equilibrium with each other (P= 2), the most abundant one consisting of A saturated with B, the minor one a trace of B saturated with A. (a between a' and a'')
- A stage is reached when so much B is present that it can dissolve all the A, and the system reverts to a single phase. The addition of more B now simply dilutes the solution, and from then on it remains a single phase. (after a'')

1. Critical solution point



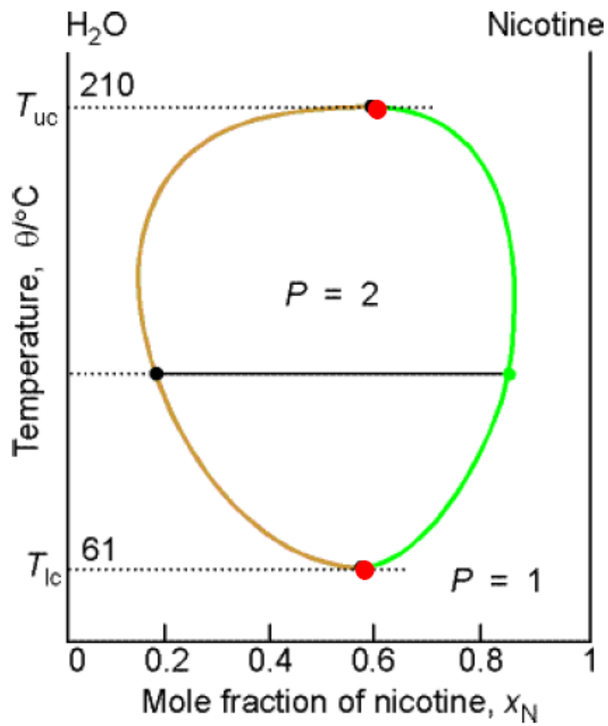
T-C diagram for hexane and nitrobenzene at 1 atm.

The upper critical solution temperature, T_{UC}, is the highest temperature at which phase separation occurs. Above T_{UC} the two components are fully miscible. This temperature exists because the greater thermal motion overcomes any potential energy advantage in molecules of one type being close together.



T-C diagram for water and triethylamine at 292K

Some systems show a lower critical solution temperature, T_{LC}, below which they mix in all proportions and above which they form two phases. In this case, at low T the two components are more miscible because they form a weak complex; at higher T the complexes break up and the two components are less miscible.

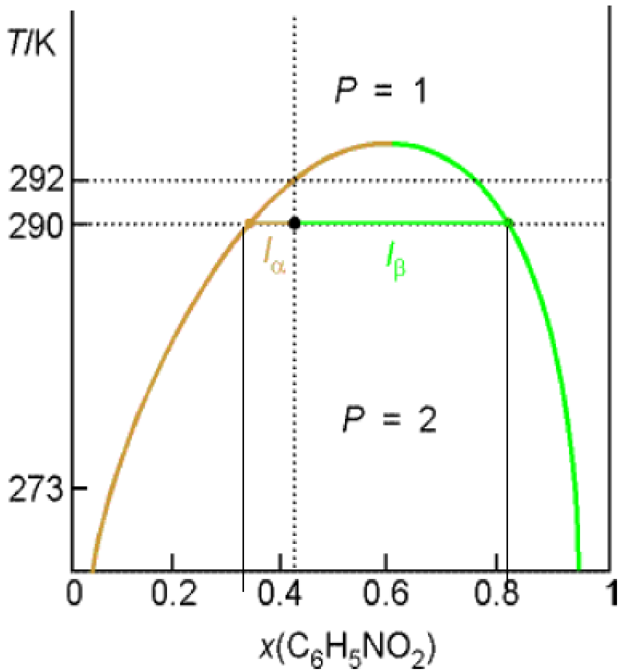


The temperature-composition diagram for water and nicotine, which has both upper and lower critical temperatures.

Some systems, like water and nicotine, have both upper and lower critical solution temperatures. They occur because, after the weak complexes have been disrupted, leading to partial miscibility, the thermal motion at higher temperatures homogenizes the mixture again, just as in the case of ordinary partially miscible liquids.

Interpreting a liquid-liquid phase diagram

A mixture of 50 g of hexane (0.59 mol C_6H_{14}) and 50 g nitrobenzene (0.41 mol $C_6H_5NO_2$) was prepared at 290 K. What are the compositions of the phases, and in what proportions do they occur? To what temperature must the sample be heated in order to obtain a single phase?



T-C diagram for hexane and nitrobenzene at 1 atm.

Answer: denote hexane by H and nitrobenzene by N;

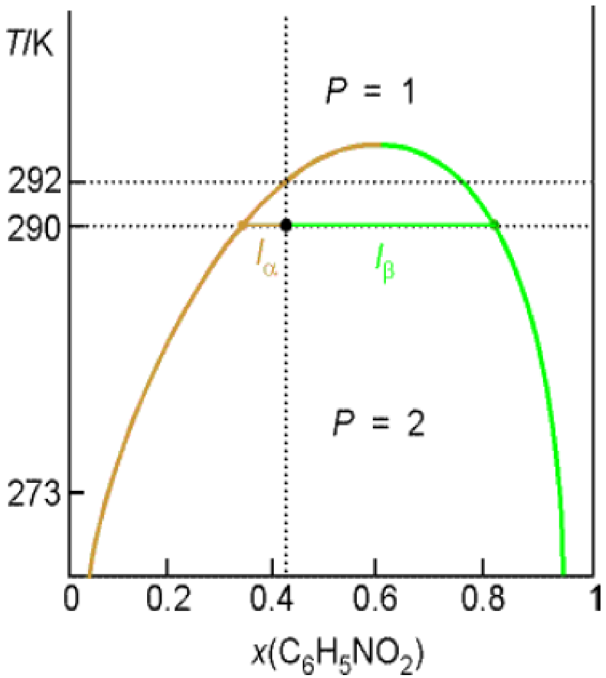
The point $x_N = 0.41$, $T = 290$ K occurs in the two-phase region of the phase diagram.

(1)The compositions of the two phases

The horizontal tie line cuts the phase boundary at

$x_N = 0.35$ and $x_N = 0.83$,

so those are the compositions of the two phases.



T-C diagram for hexane and nitrobenzene at 1 atm.

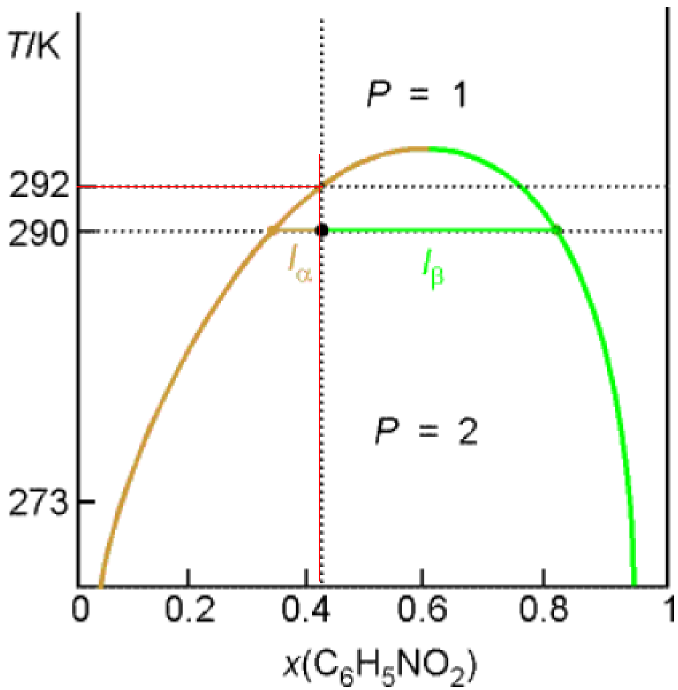
(2). The proportion of the mixture at *T*=290K

$$n_\alpha l_\alpha = n_\beta l_\beta$$

so the ratio of the amounts of each phase is equal to the ratio of the distances l_α and l_β

l_α = 0.41-0.35; l_β = 0.83-0.41

$$\frac{n_\alpha}{n_\beta} = \frac{l_\beta}{l_\alpha} = \frac{0.83-0.41}{0.41-0.35} = 7$$



T-C diagram for hexane and nitrobenzene at 1 atm.

(3). to *T* obtain a single phase

Heating the sample to

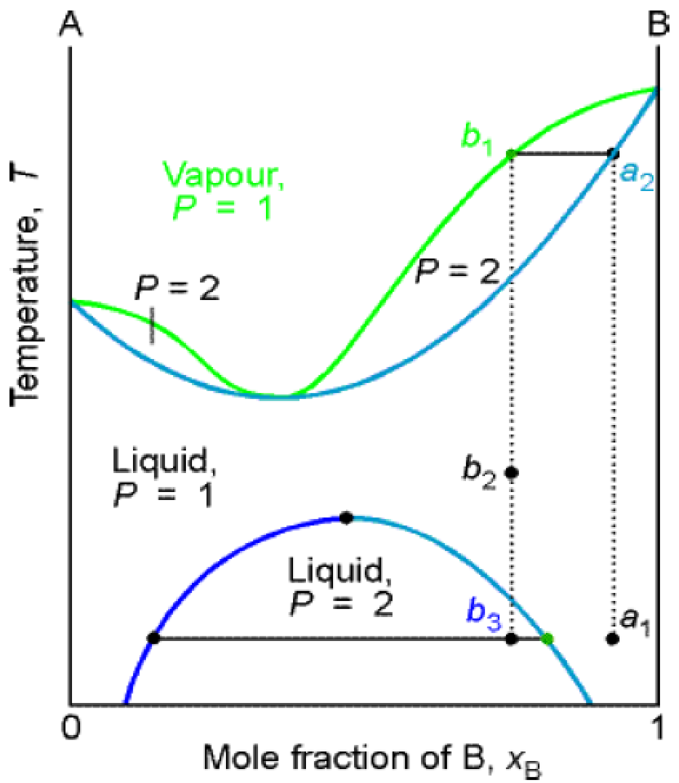
T=292 K

takes into the single-phase region.

2. The distillation of partially miscible liquids

A pair of liquids that are partially miscible and form a low-boiling azeotrope is common combination because both properties reflect the tendency of the two kinds of molecule to avoid each other. There are two possibilities: one in which the liquids become fully miscible before they boil, the other in which boiling occurs before mixing is complete.

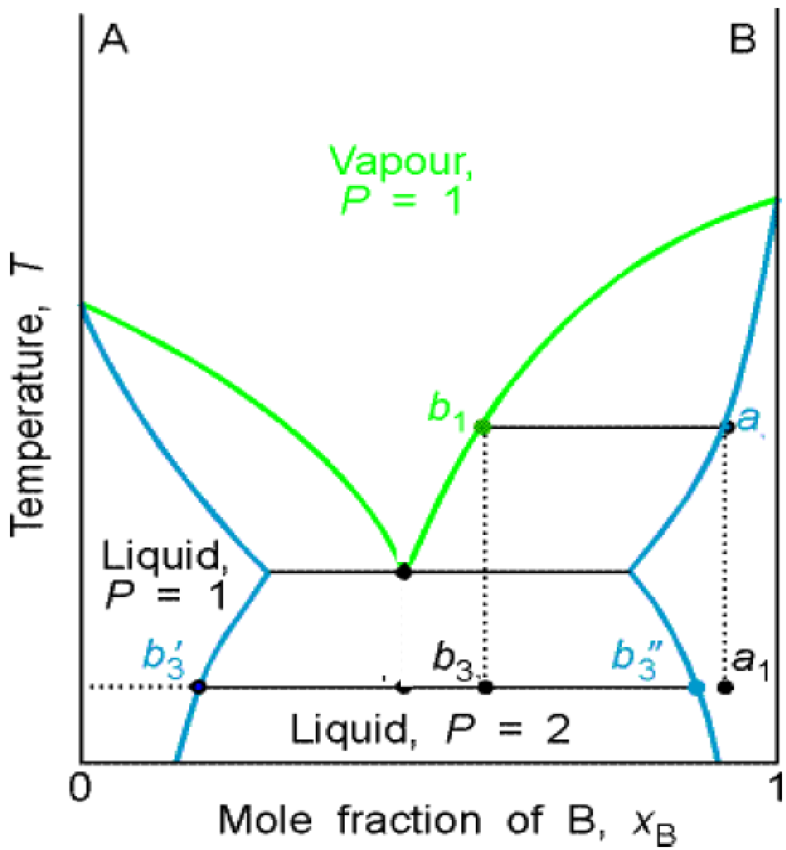
The phase diagram for two components that become fully miscible before they boil.



The temperature-composition diagram for a binary system in which the T_{uc} is less than the boiling point at all compositions

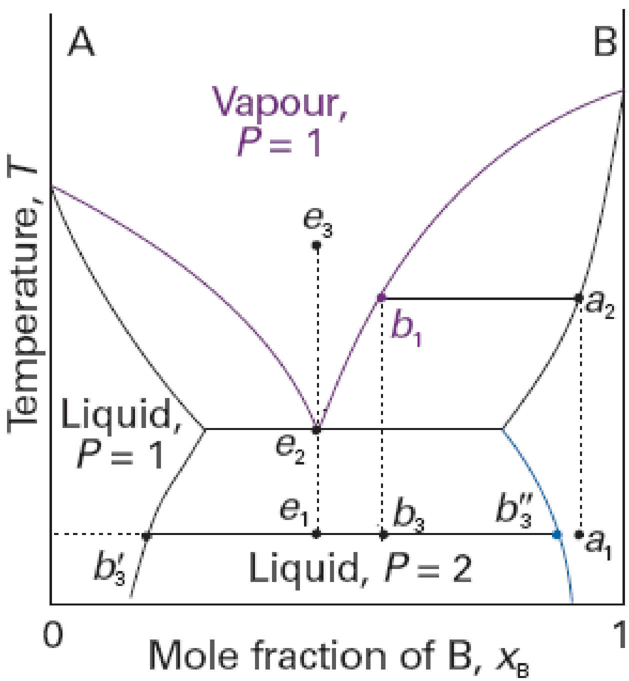
Distillation of a mixture of composition a_1, leads to a vapour of composition b_1, which condenses to the completely miscible single-phase solution at b_2.Phase separation occurs only when cooled to a point in the two-phase liquid region, such as b_3.

There is no upper critical solution temperature.



The temperature-composition diagram for a binary system in which boiling occurs before the two liquids are fully miscible.

The distillate obtained from a liquid initially of composition a_1 has composition b_3 and is a two-phase mixture. One phase has composition b_3' and the other has composition b_3''.



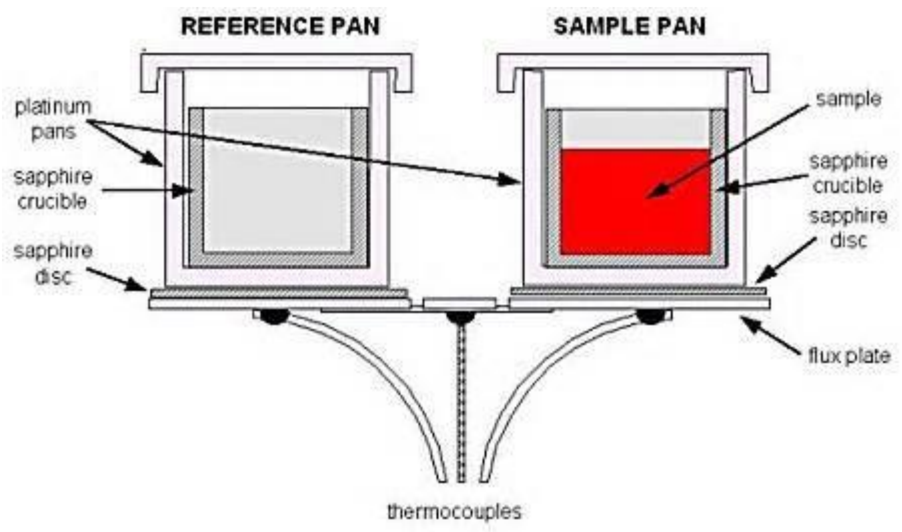
The behaviour of a system of composition represented by the isopleth e is interesting. A system at e_1 forms two phases, which persist (but with changing proportions) up to the boiling point at e_2. The vapour of this mixture has the same composition as the liquid (the liquid is an zeotrope). Similarly, condensing a vapour of composition e_3 gives a two-phase liquid of the same overall composition. At a fixed temperature, the mixture vaporizes and condenses like a single substance.

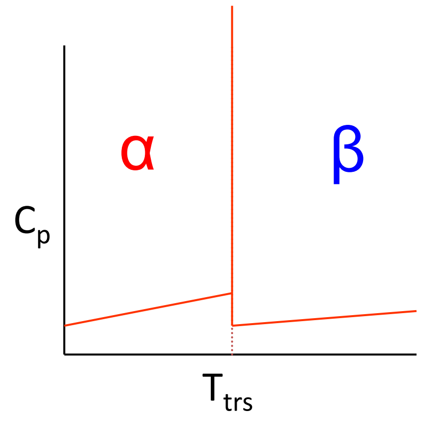
6.6 Liquid-solid phase diagrams

1. Thermal analysis

Differential scanning calorimetry (DSC)

- When the sample undergoes a physical transformation such as phase transitions, more or less heat will need to flow to it than the reference to maintain both at the same temperature.
- Whether less or more heat must flow to the sample depends on whether the process is exothermic or endothermic.





Cooling curve

- Used to determine phase transition temperature
- Record T of material vs time, as it cools from its molten state through solidification and finally to RT (at a constant pressure)

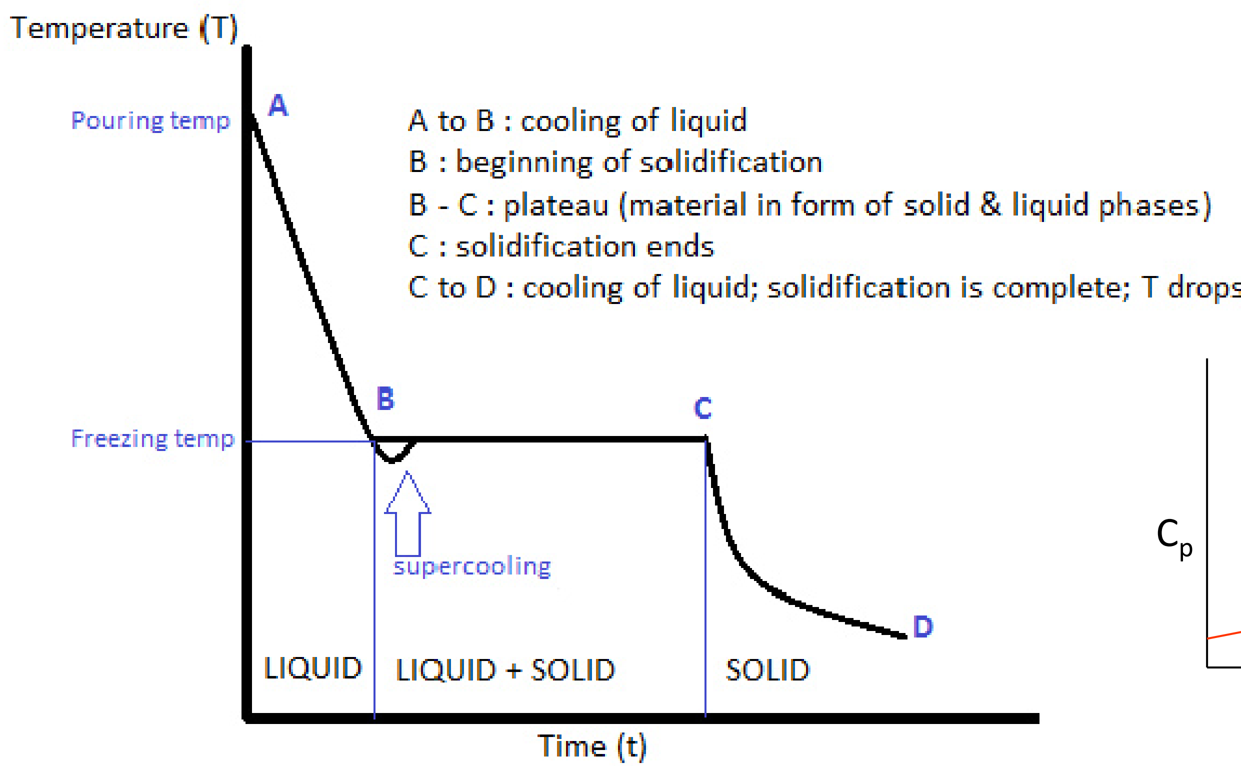
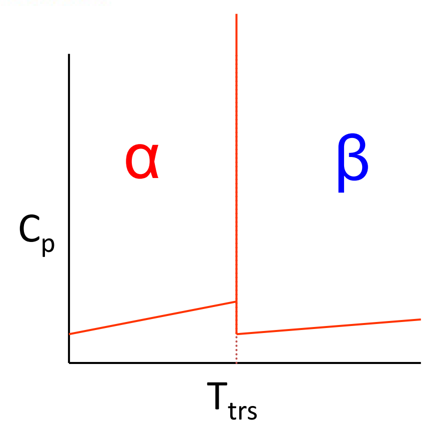
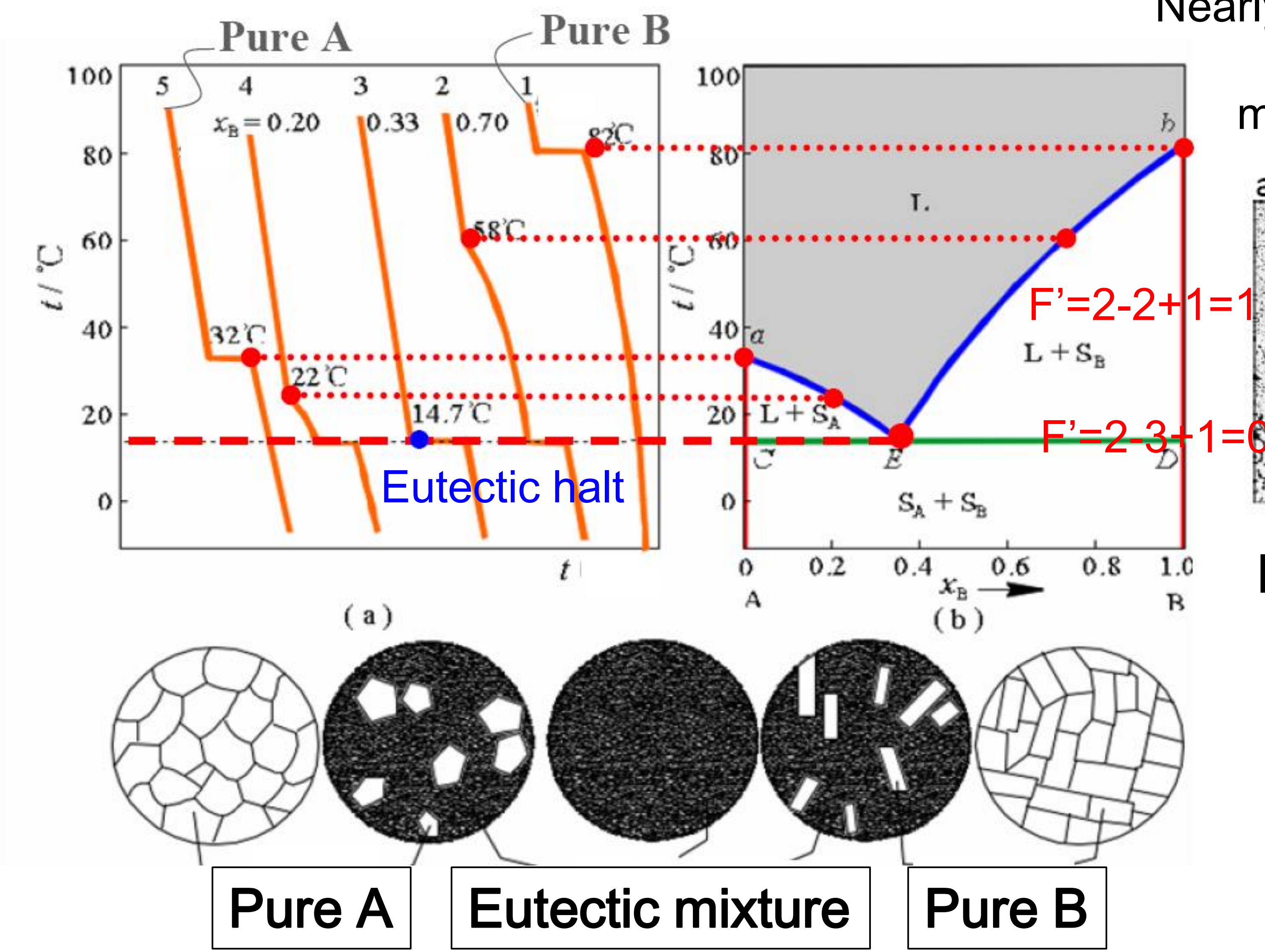


Diagram showing cooling curve of a pure metal

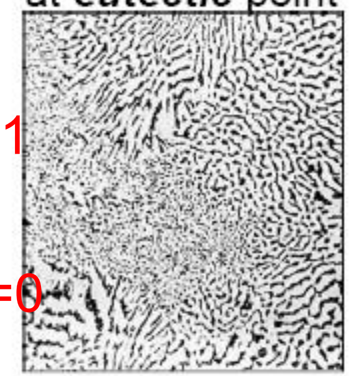


2. Binary eutectic alloy system



Nearly homogeneous mixture of microcrystals

at *eutectic* point



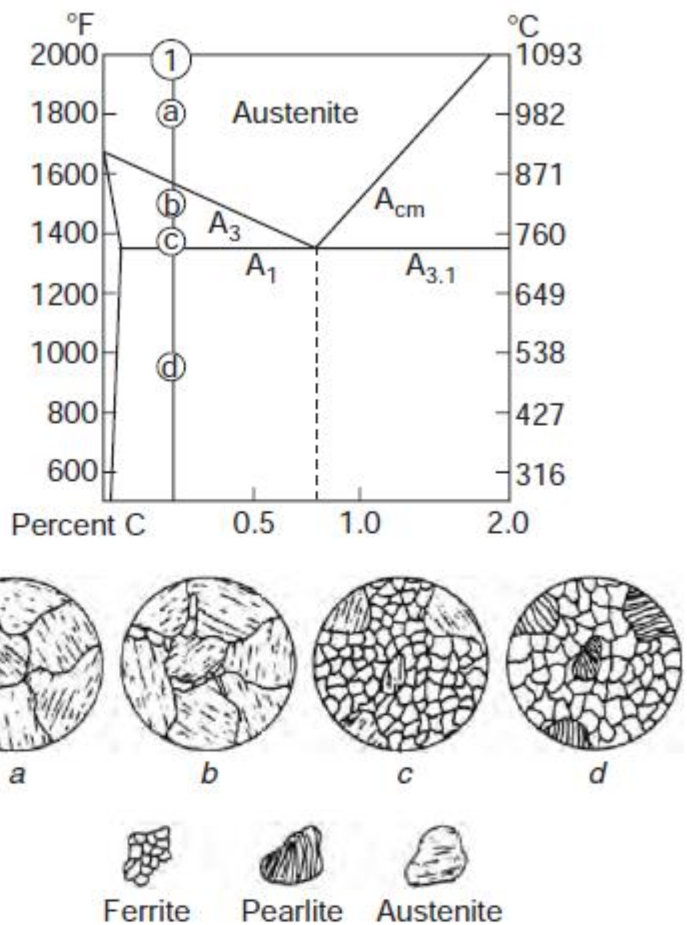
Microscopy

X-ray

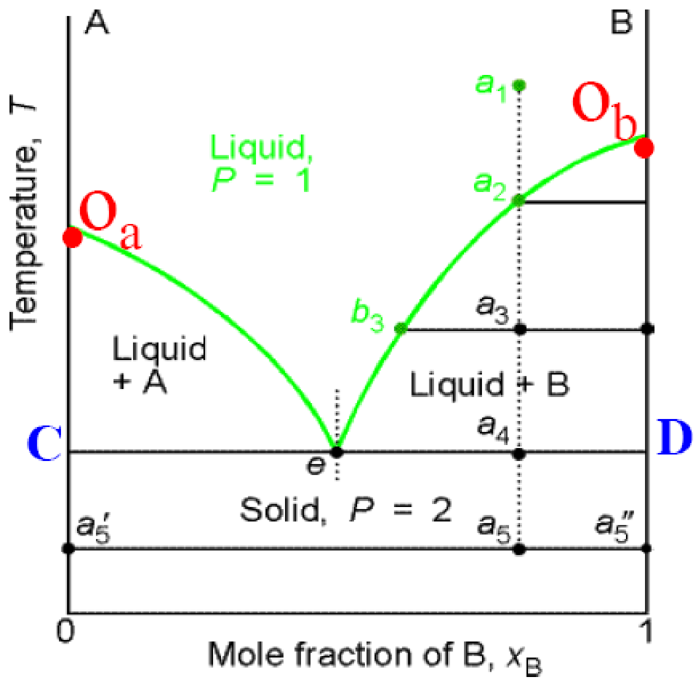
• Hypoeutectoid steel

Compositions to the left of eutectoid (< 0.83 wt % C) alloys.

$$\gamma \rightarrow \alpha + Pearlite$$



- Austenite steel is slowly cooled.
- Ferrite begins to form along the austenite grain boundaries.
- Further cooling results in a ferrite-rich phase, with some remaining austenite crystals.
- At the eutectoid point of 723◦C, the residual austenite is converted to pearlite, yielding a state that contains both **ferrite and pearlite** crystals upon further cooling.



The temperature-composition phase diagram for two almost immiscible solids and their completely miscible liquids.

The isopleth through *e* corresponds to the eutectic composition, the mixture with lowest melting point.

Points, lines, and regions

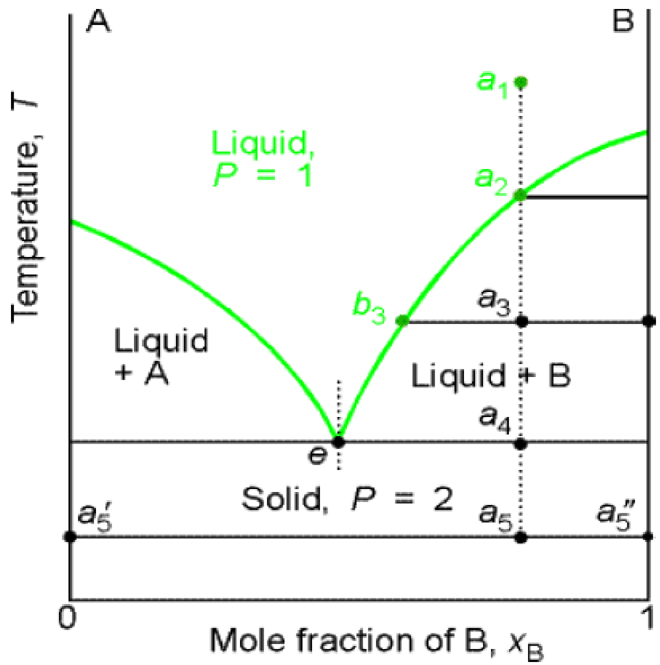
O_a: freezing point of pure *a*

O_b: freezing point of pure *b*

O_a–*e*, O_b–*e* :liquid phase line

O_a–C, O_b–D :solid phase line

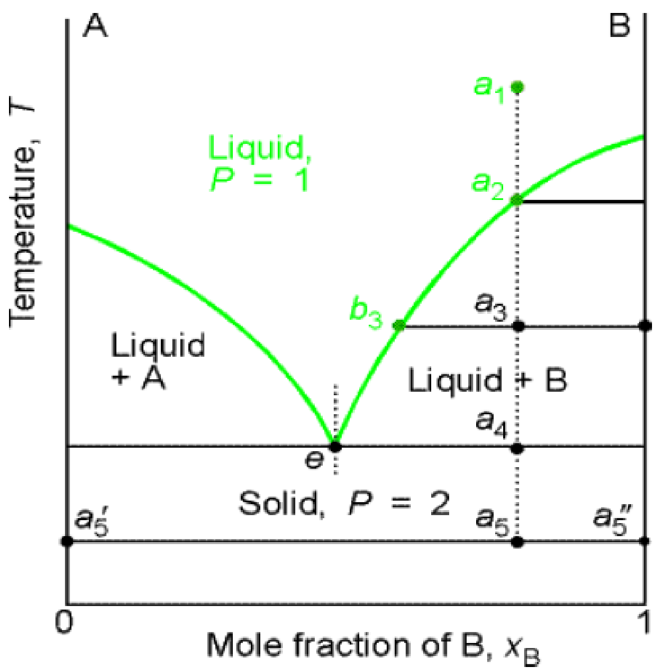
C–*e*–D: three phase line



The temperature-composition phase diagram for two almost immiscible solids and their completely miscible liquids.

(1) $a_1 \rightarrow a_2$

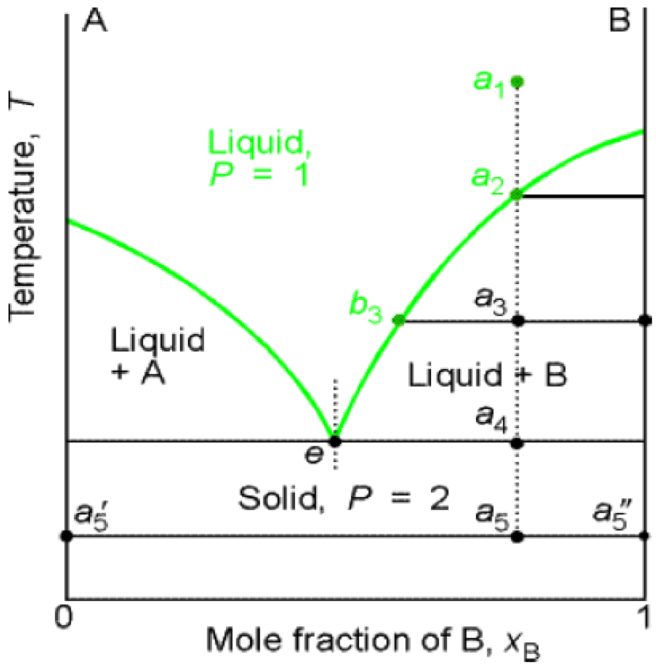
The system enters the two-phase region labelled Liquid + B. Pure solid B begins to come out of solution and the remaining liquid becomes richer in A.



The temperature-composition phase diagram for two almost immiscible solids and their completely miscible liquids.

(1) $a_2 \rightarrow a_3$

More of the solid forms. The relative amounts of the solid and liquid are given by the lever rule. At this stage there are roughly equal amounts of each. The liquid phase is richer in A than before because some B has been deposited.



The temperature-composition phase diagram for two almost immiscible solids and their completely miscible liquids.

(1) $a_3 \rightarrow a_4$

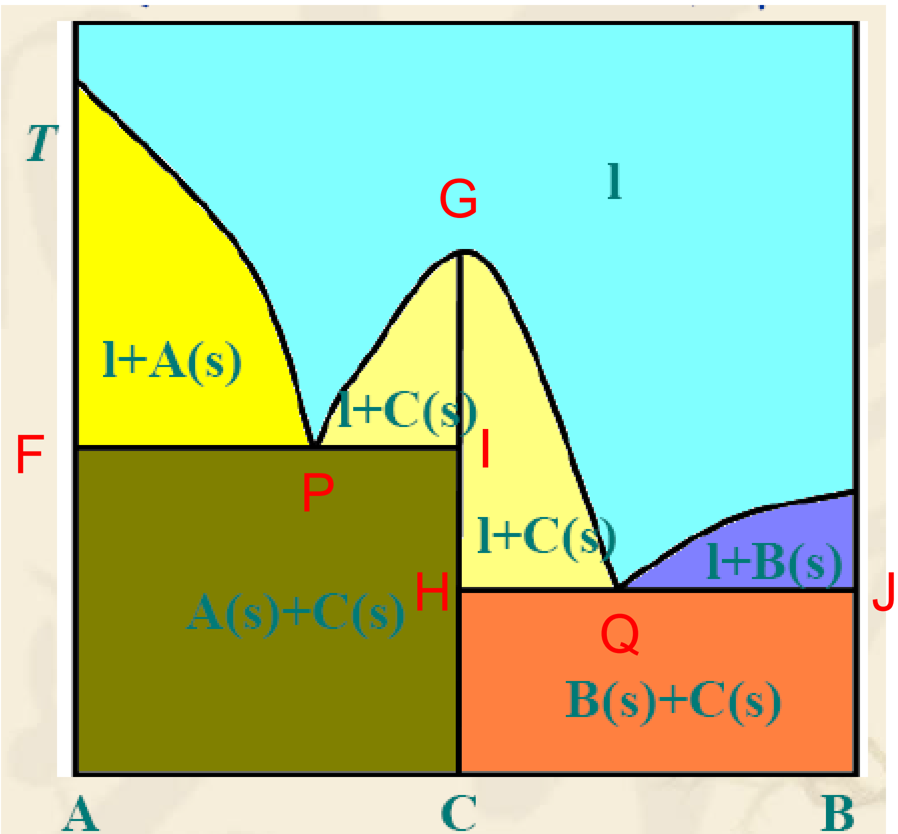
At the end of this step, there are three phases: liquid, solid B and solid A. The liquid composition is given by e. This liquid will freez to give a two-phase system of pure B and pure A.

3. Reacting systems

Many binary mixtures react to produce compounds. We shall illustrate some of the principles involved with a system the forms a compound C that also forms eutectic mixtures with the species A and B, respectively.

$A + B = C$ $(stable\ both\ in\ the\ liquid\ and\ solid\ phase)$

C=AB



Two triple lines:

FI: P=3，(l、A、C）,F=0

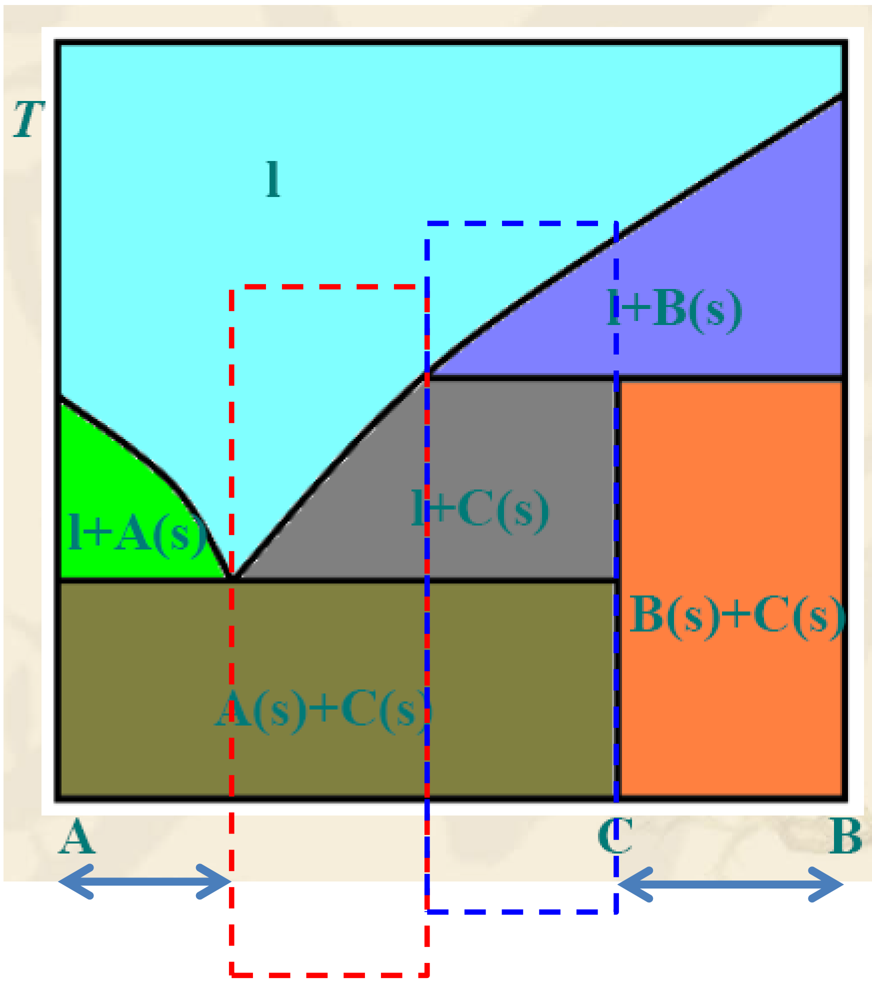
JH: P=3，(l、C、B）,F=0

P: Eutectic mixture of A and C

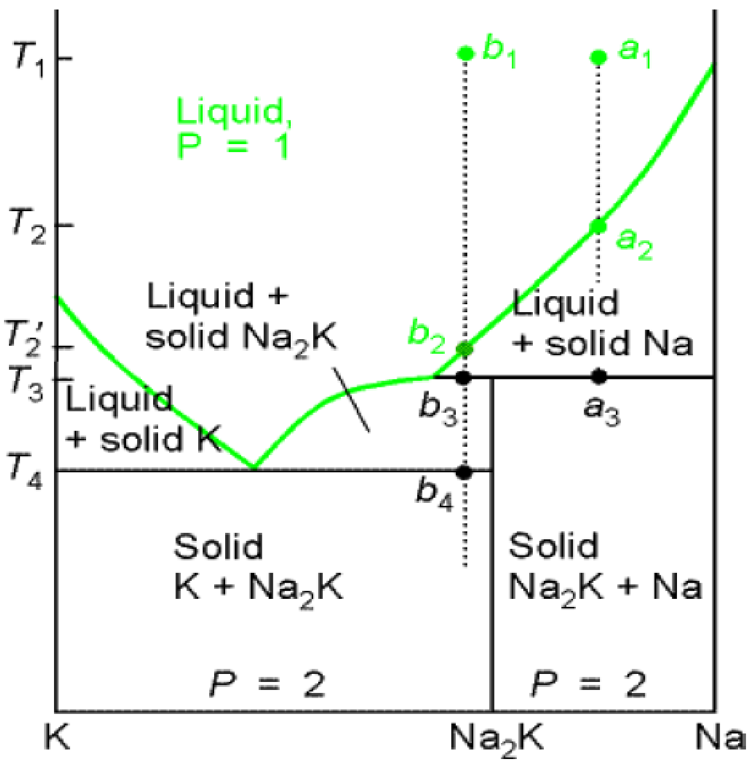
Q: Eutectic mixture of B and C

G: the melt point of compound C

4. Incongruent melting



In some cases the compound C is not stable as a liquid. The Compound C has an incongruent melting, in which the compound C melts into its components and does not itself form a liquid phase.

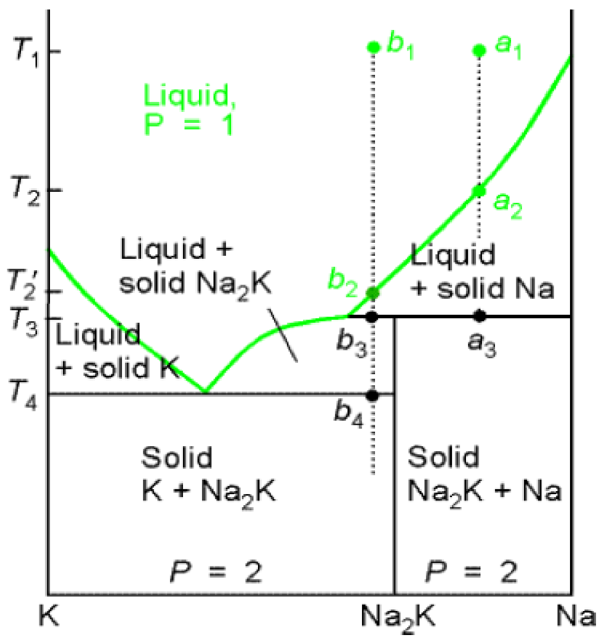


The phase diagram for an actual system

In some cases the compound C is not stable as a liquid. An example is the alloy Na_2K that a liquid at a_1 is cooled:

(1) $a_1 \rightarrow a_2$. Some solid Na is deposited, and the remaining liquid is richer in K.

(2) $a_1 \rightarrow$ Just below a_3. The sample is now entirely solid, and consists of solid Na and solid Na_2K.

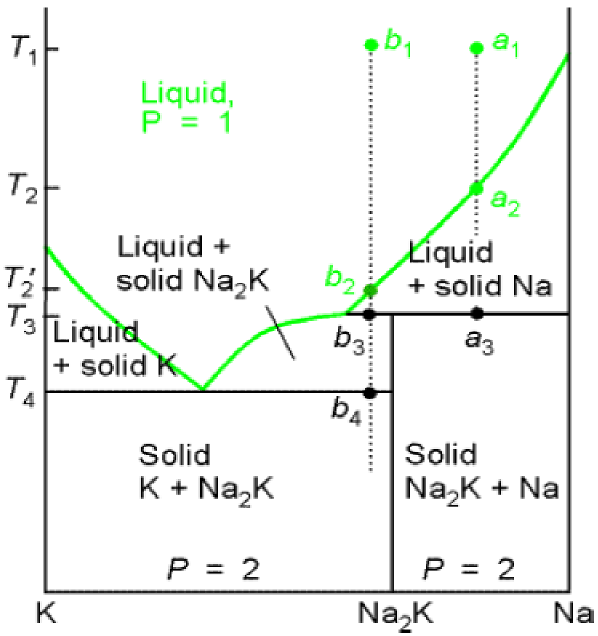


The phase diagram for an actual system

The isopleth through b_1

(1) $b_1 \rightarrow b_2$. No obvious change occurs until the phase boundary is reached at b_2 when solid Na begins to deposit.

(2) $b_2 \rightarrow b_3$. Solid Na deposits, but at b_3 a reaction occurs to form Na_2K: this compound is formed by the K atoms diffusing into the solid Na. At this stage the liquid Na/K mixture is in equilibrium with a little solid Na_2K, but there is still no liquid compound of Na_2K.



The phase diagram for an actual system

(3) $b_3 \rightarrow b_4$. As cooling continues, the amount of solid compound increases until at b_4 the liquid reaches its eutectic composition. It then solidifies to give a two-phase solid consisting of solid K and solid Na_2K.

If the solid is reheated, the sequence of events is reversed. No liquid Na_2K forms at any stage because it is too unstable to exist as a liquid. This behaviour is an example of incongruent melting, in which a compound melts into its components and does not itself form a liquid phase.

Summary

1. The phase rule

$$F = C - P + 2$$

J.W. Gibbs deduced the phase rule:

F: the degree of freedom or variance;

C: the number of components;

P: the number of phases at equilibrium.

2. Two-component phase diagrams

- Miscible ideal solutions
  - Vapour pressure-composition diagrams
  - Temperature-compostion diagrams
  - The Level rule on the tie line
- Miscible real solution
  - Azeotropic point and azeotrope
  - Distillation
- Immiscible liquids
- Partially miscible liquid-liquid diagrams
  - Critical point
- Liquid-solid phase diagrams
  - The cooling curves
  - Eutectics